Astrea Academy Trust
INSPIRING BEYOND MEASURE

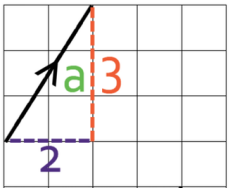
Name .....

Class .....

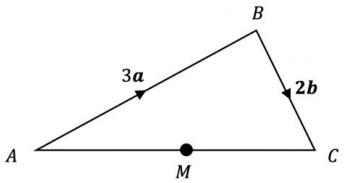
Tutor Group .....

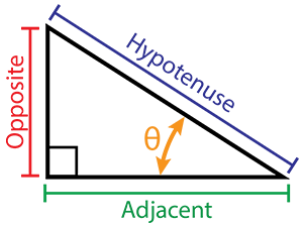
YEAR 11 CORE MATHEMATICS

Booklet 3 – Vectors & Trigonometry



$a = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$





KEY VOCABULARY LIST	
Direction	Where something is pointing.
Magnitude	The size of something. The magnitude of a number (also called its absolute value) is its distance from zero, so • the magnitude of 6 is 6 • the magnitude of −6 is also 6 The magnitude of a vector is its length (ignoring direction).
Vector	A vector quantity has both size and direction. Vectors can be added, subtracted and multiplied by a scalar. Geometrical problems can be solved using vectors.
Column Vector	Notation showing horizontal and vertical movement of a point: $\begin{pmatrix} x \\ y \end{pmatrix}$
Resultant vector	A resultant vector is the sum of two or more vectors. It is the result of adding two or more vectors together.
Scalar	A number on its own (used when dealing with vectors or matrices).
Parallel	Two lines are parallel if they never meet, maintaining a constant distance between them.
Trigonometry	The study of triangles.
Hypotenuse	The longest side in a right-angled triangle, opposite the right angle.
Adjacent side	The side next to the angle of interest (not the hypotenuse).
Opposite side	The side opposite the angle of interest.

CONTENTS		PAGE
SUBTOPIC 1	**COLUMN VECTORS**	**3**
SUBTOPIC 1A	Definition & Notation	3
SUBTOPIC 1B	Multiplying Vectors	7
SUBTOPIC 1C	Adding and Subtracting Vectors	10
SUBTOPIC 2	**PARALLEL VECTORS**	**16**
SUBTOPIC 3	**VECTOR GEOMETRY**	**19**
	EXAM STYLE QUESTIONS	**22**
SUBTOPIC 4	**RIGHT-ANGLED TRIGONOMETRY**	**24**
SUBTOPIC 4A	Trigonometry - Building Blocks	24
SUBTOPIC 4B	Calculating a missing length	35
SUBTOPIC 4C	Calculating a missing angle	39
SUBTOPIC 4D	Applications of Trigonometry	43
	GCSE EXAM QUESTIONS	50
SUBTOPIC 5	**PROBLEMS WITH TRIGONOMETRY & PYTHAGORAS' THEOREM**	**51**
SUBTOPIC 5A	Pythagoras' Theorem - RECAP	51
SUBTOPIC 5B	Choosing between Pythagoras and Trigonometry	55
SUBTOPIC 5C	Solving Problems with Pythagoras and Trigonometry	59
	GCSE EXAM QUESTIONS	60
SUBTOPIC 5D	Bearings with Pythagoras' Theorem & Trigonometry	61
SUBTOPIC 6	**EXACT TRIGONOMETRIC VALUES**	**64**
	EXAM STYLE QUESTIONS	**65**

BOOKLET STRUCTURE

Start of subtopic:

- Read and annotate the information in the Frayer model or introduction text
- I do (teacher led example).
- We do (teacher and scholar led example).
- You do (independent scholar example).
- Whiteboard check.
- Independent practice and mixed practice.

End of subtopic:

- Skills checks – key questions to help your teachers to pinpoint any issues.
- GCSE questions – to ensure you are confident with the style you will encounter in your exams.

Subtopic 1: Column Vectors

LEARNING OUTCOME: To understand and use column vector notation.

Objectives:
- Interpret vectors as a movement from one position to another.
- Use column notation to identify, describe and draw vectors on a grid.
- Use vector geometry to find pathways between points using vectors.

Keywords:
- Vector
- Scalar
- Magnitude
- Direction

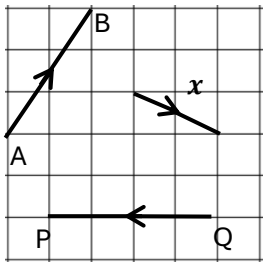
Vector

Definition:
A quantity with both direction and magnitude.

It describes the movement from one position to another.

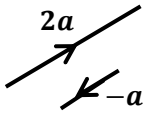
Characteristics:
Represented graphically by a line of a given length with an arrow to show direction.
Can be written using one letter notation, $\boldsymbol{a}$, two letter notation, $\overrightarrow{AB}$, or a column vector $\begin{pmatrix} x \\ y \end{pmatrix}$
Can be scaled, added and subtracted.

Examples:

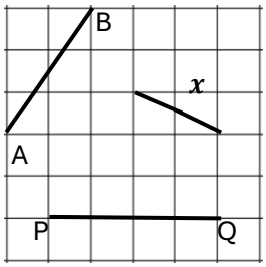


$\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ $\overrightarrow{AB}$

$\begin{pmatrix} -3 \\ 7 \end{pmatrix}$

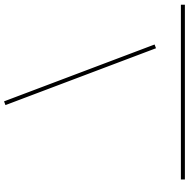


Non-examples:



$(2,3)$ AB

$\begin{pmatrix} \frac{2}{3} \end{pmatrix}$



Subtopic 1A: Definition & Notation

	Example 1. I do...	*We do...*	*You do...*
Horizontal Movement +: Right −: Left $\begin{pmatrix} x \\ y \end{pmatrix}$ $\begin{pmatrix} \leftrightarrow \\ \updownarrow \end{pmatrix}$ Vertical Movement +: Up −: Down	Write as a column vector: a) 2 right, 5 up b) 2 left, 5 up Write in words: $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$	Write as a column vector: a) 4 left, 3 up b) 10 right Write in words: $\begin{pmatrix} -6 \\ 1 \end{pmatrix}$	Write as a column vector: a) 7 left, 5 down b) 3 down Write in words: $\begin{pmatrix} -8 \\ 0 \end{pmatrix}$

Independent Practice

Question 1. Fill the gaps to give matching descriptions of the column vector & worded directions:

$a)$ $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$	6 _____ 3 _____	$b)$ $\begin{pmatrix} \square \\ 3 \end{pmatrix}$	6 left 3 _____
$c)$ $\begin{pmatrix} \square \\ \square \end{pmatrix}$	8 right 7 down	$d)$ $\begin{pmatrix} \square \\ 0 \end{pmatrix}$	3 left __ _____

Question 2. Write as a column vector:		**Question 3.** Write in words:	
a) 8 right 3 down	b) 9 left 2 up	$a)\ \begin{pmatrix}-2\\8\end{pmatrix}$	$b)\ \begin{pmatrix}3\\4\end{pmatrix}$
c) 6 down 7 right	d) 4 left	$c)\ \begin{pmatrix}0\\5\end{pmatrix}$	$d)\ \begin{pmatrix}-10\\-2\end{pmatrix}$

Question 4. Circle 'true' or 'false' for each of the following;

1) $\begin{pmatrix}4\\2\end{pmatrix}$ = →4↑2 True/False

2) $\begin{pmatrix}3\\2\end{pmatrix}$ = →3↑2 True/False

3) $\begin{pmatrix}-3\\2\end{pmatrix}$ = →3↑2 True/False

4) $\begin{pmatrix}-4\\2\end{pmatrix}$ = ←4↑2 True/False

5) $\begin{pmatrix}-1\\-1\end{pmatrix}$ = ←1↑1 True/False

6) $\begin{pmatrix}-2\\-2\end{pmatrix}$ = ←2↓2 True/False

7) $\begin{pmatrix}-3\\-3\end{pmatrix}$ = ←3↓3 True/False

8) $\begin{pmatrix}3\\-1\end{pmatrix}$ = →3↓1 True/False

9) $\begin{pmatrix}-3\\-1\end{pmatrix}$ = ←3↓1 True/False

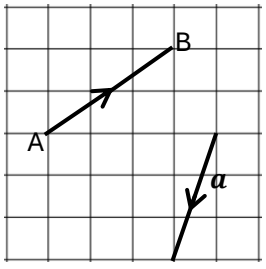
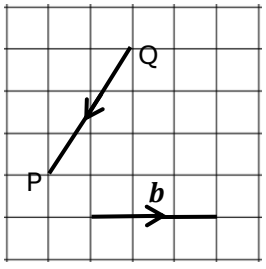
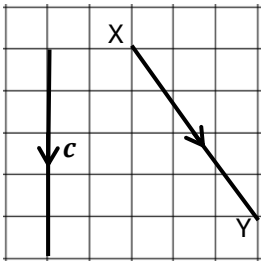
10) $\begin{pmatrix}4\\-1\end{pmatrix}$ = →4↑1 True/False

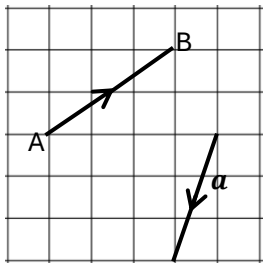
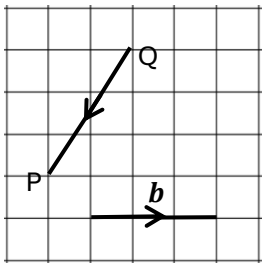
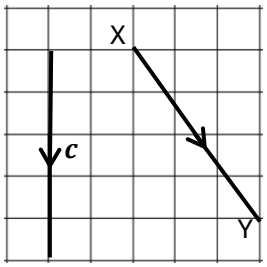
11) $\begin{pmatrix}5\\-1\end{pmatrix}$ = →5↓1 True/False

12) $\begin{pmatrix}6\\-1\end{pmatrix}$ = →6↑1 True/False

13) $\begin{pmatrix}7\\-1\end{pmatrix}$ = →7↓1 True/False

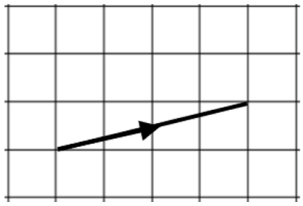
14) $\begin{pmatrix}-8\\0\end{pmatrix}$ = ←8↕0 True/False

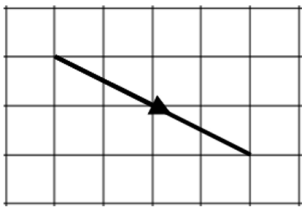
Steps	***Example 2. I do...***	***We do...***	***You do...***
1. Use the arrow to determine the start point of the vector. 2. Count the number of horizontal and vertical squares between the start and end points. 3. Write as a column vector.	Write the vectors below as a column vectors.  	Write the vectors below as a column vectors.  	Write the vectors below as a column vectors. 

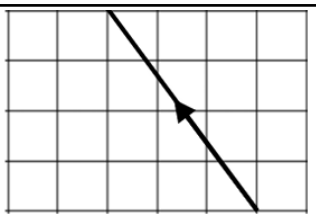
Steps	*Example 3. I do...*	*We do...*	*You do...*
1. Use the arrow to determine the start point of the vector. 2. Count the number of horizontal and vertical squares between the start and end points. 3. Write as a column vector.	Write the vectors below as a column vectors.  	Write the vectors below as a column vectors.  	Write the vectors below as a column vectors. 

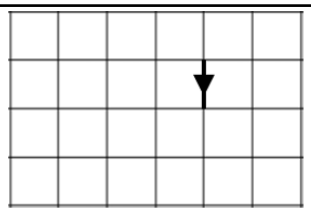
Independent Practice

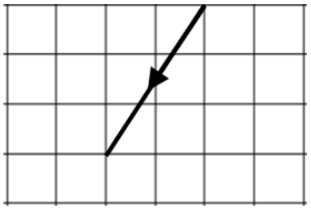
Question 5. Write as a column vector:

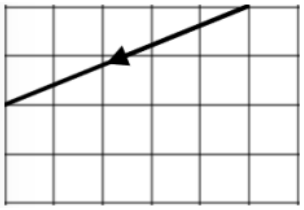
a) 

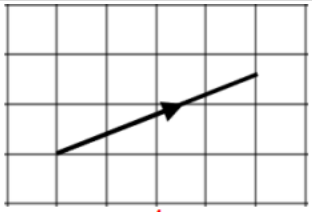
b) 

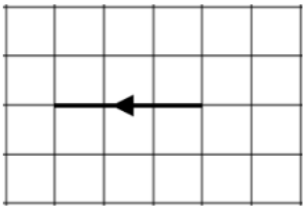
c) 

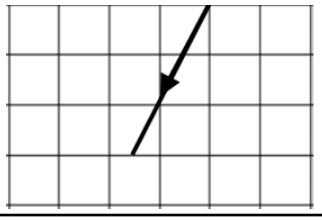
d) 

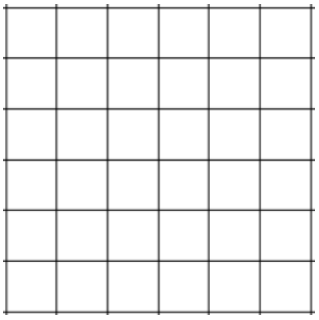
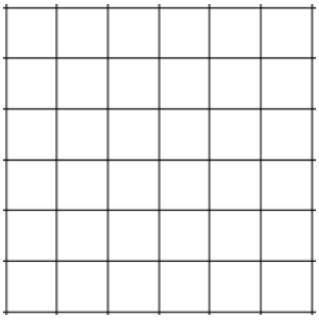
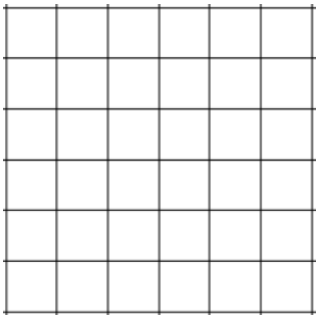
e) 

f) 

g) 

h) 

i) 

Steps	*Example 4. I do...*	*We do...*	*You do...*
1. Choose the start point of the vector. 2. Count the number of horizontal and vertical squares in the vector. 3. Draw in the end point. 4. Join the start and end points and draw the arrow to show the direction.	Draw the vectors: $\boldsymbol{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \boldsymbol{b} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ 	Draw the vectors: $\boldsymbol{c} = \begin{pmatrix} -3 \\ -2 \end{pmatrix}, \boldsymbol{d} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$ 	Draw the vectors: $\boldsymbol{e} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}, \boldsymbol{f} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$ 

Question 6. Draw the following column vectors:

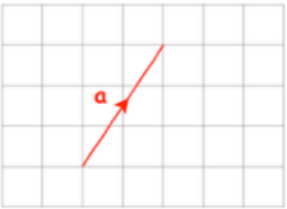
$\begin{pmatrix}5\\3\end{pmatrix}$ $\begin{pmatrix}0\\6\end{pmatrix}$ $\begin{pmatrix}2\\-3\end{pmatrix}$

$\begin{pmatrix}-4\\0\end{pmatrix}$ $\begin{pmatrix}-1\\-5\end{pmatrix}$ $\begin{pmatrix}0.5\\-2\end{pmatrix}$

Question 7.

Mark has been asked to draw the vector $\boldsymbol{a} = \begin{pmatrix}3\\2\end{pmatrix}$.

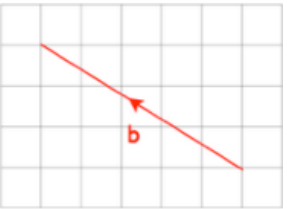
Explain his mistake.



Question 8.

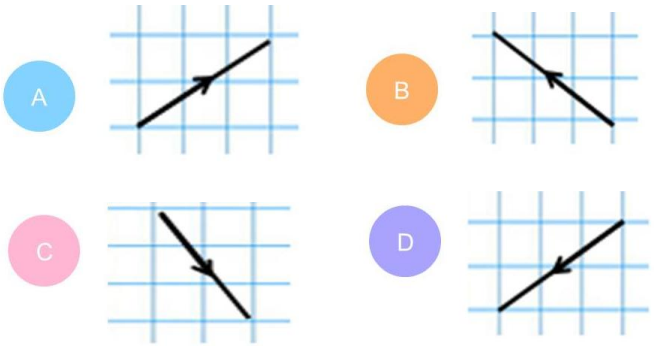
Abby has been asked to draw the vector $\boldsymbol{b} = \begin{pmatrix}5\\-3\end{pmatrix}$.

What mistake(s) has she made?

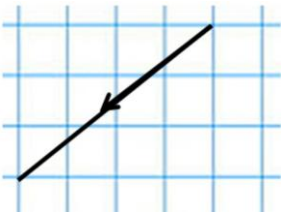


Skills Check

Which diagram shows the column vector $\begin{pmatrix}-3\\2\end{pmatrix}$?

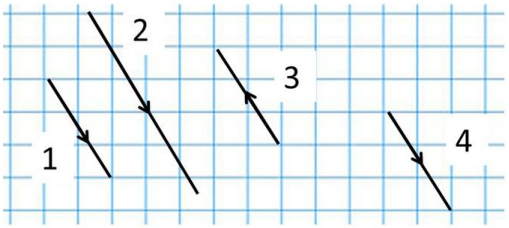


Which column vector matches the diagram?



A $\begin{pmatrix}4\\3\end{pmatrix}$ B $\begin{pmatrix}-3\\-4\end{pmatrix}$

C $\begin{pmatrix}4\\-3\end{pmatrix}$ D $\begin{pmatrix}-4\\-3\end{pmatrix}$



Which vectors in the diagram are the same?

A 1 and 4

B All

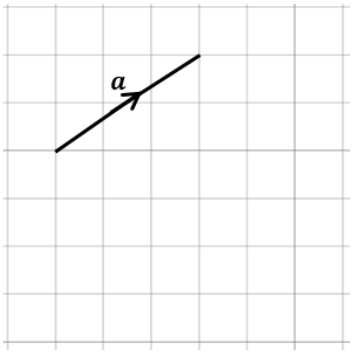
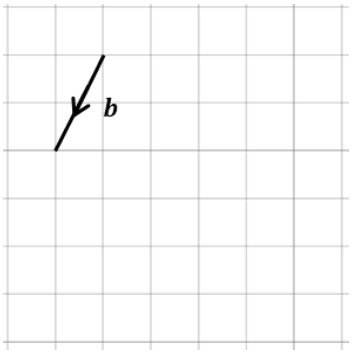
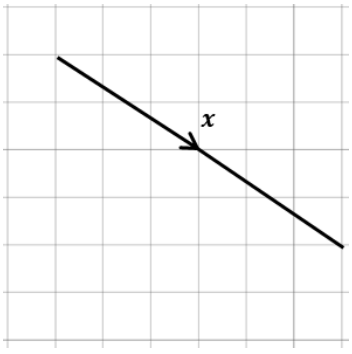
C 1, 3 and 4

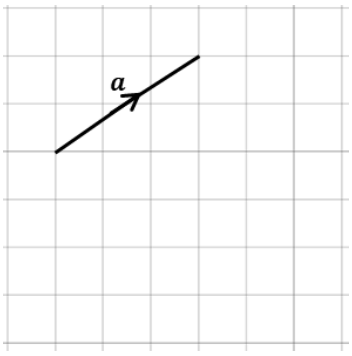
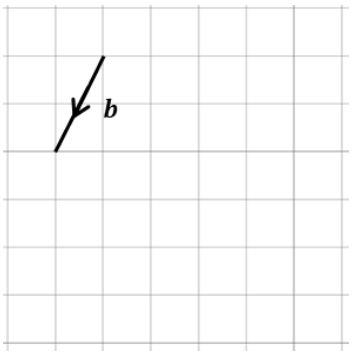
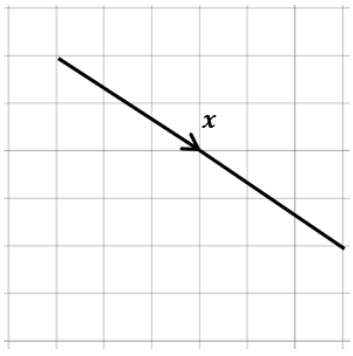
D None

Subtopic 1B: Multiplying Vectors

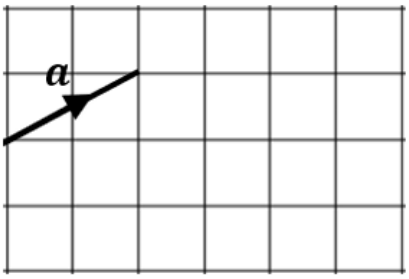
1. A **scalar** is a quantity with **only magnitude and no direction**. They are described by a **single**
2. **number**, and they often represent something measurable such as a length or a time
3. e.g. a measurement of 5m or a time of 24 seconds.
4. We can also **multiply vectors by a scalar**, to change the **magnitude;** make it longer or shorter.

5. If we multiply a vector by a **positive** scalar, the **direction** is **unchanged.**
6. If we multiply a vector by a **negative** scalar, the **direction** is **reversed**.

Steps	*Example 5. I do...*	*We do...*	*You do...*
1. Count the horizontal and vertical movement for the vector. 2. Pick a new start point and repeat this the required number of times.	The vector $\boldsymbol{a}$ is shown below. Draw the vector $2\boldsymbol{a}$. 	The vector $\boldsymbol{b}$ is shown below. Draw the vector $3\boldsymbol{b}$. 	The vector $\boldsymbol{x}$ is shown below. Draw the vector $\frac{1}{2}\boldsymbol{x}$. 
	Write as column vectors:	Write as column vectors:	Write as column vectors:

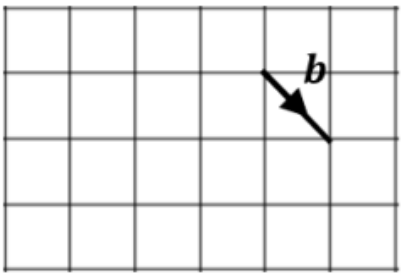
Steps	*Example 6. I do...*	*We do...*	*You do...*
1. Count the horizontal and vertical movement for the vector. 2. Pick a new start point and repeat this the required number of times in the opposite direction.	The vector $\boldsymbol{a}$ is shown below. Draw the vector $-\boldsymbol{a}$. 	The vector $\boldsymbol{b}$ is shown below. Draw the vector $-3\boldsymbol{b}$. 	The vector $\boldsymbol{x}$ is shown below. Draw the vector $-\frac{1}{2}\boldsymbol{x}$. 
	Write as column vectors:	Write as column vectors:	Write as column vectors:

Q9. Vector $\boldsymbol{a}$ is shown.
Draw vector $2\boldsymbol{a}$.



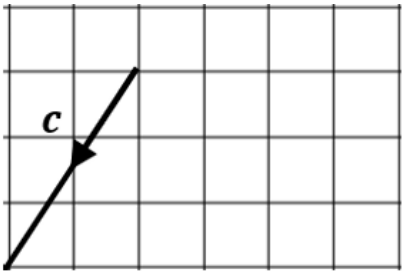
Write as column vectors:
$\boldsymbol{a} =$ $\qquad$ $2\boldsymbol{a} =$

Q10. Vector $\boldsymbol{b}$ is shown.
Draw vector $3\boldsymbol{b}$.



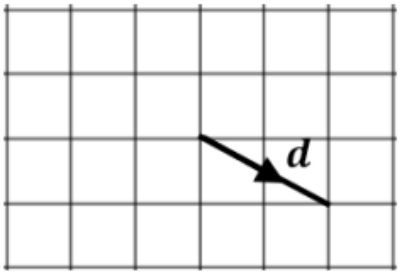
Write as column vectors:
$\boldsymbol{b} =$ $\qquad$ $3\boldsymbol{b} =$

Q11. Vector $\boldsymbol{c}$ is shown.
Draw vector $-\boldsymbol{c}$.



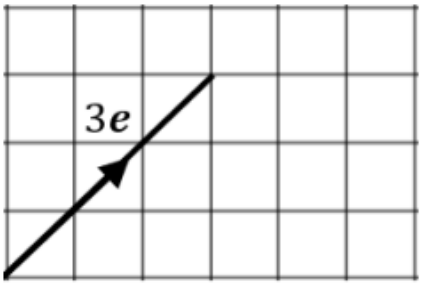
Write as column vectors:
$\boldsymbol{c} =$ $\qquad$ $-\boldsymbol{c} =$

Q12. Vector $\boldsymbol{d}$ is shown.
Draw vector $-2\boldsymbol{d}$.



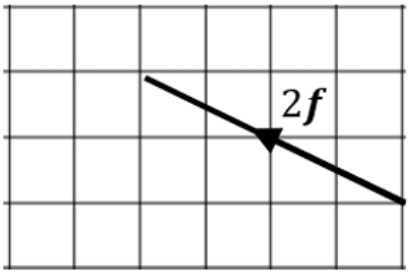
Write as column vectors:
$\boldsymbol{d} =$ $\qquad$ $-2\boldsymbol{d} =$

Q13. Vector $3\boldsymbol{e}$ is shown.
Draw vector $2\boldsymbol{e}$.



Write as column vectors:
$3\boldsymbol{e} =$ $\qquad$ $2\boldsymbol{e} =$

Q14. Vector $2\boldsymbol{f}$ is shown.
Draw vector $-\boldsymbol{f}$.



Write as column vectors:
$2\boldsymbol{f} =$ $\qquad$ $-\boldsymbol{f} =$

Steps	*Example 7. I do...*	*We do...*	*You do...*
1. Multiply both the horizontal and vertical components by the coefficient.	$\boldsymbol{a} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$ Find: 1) $4\boldsymbol{a}$ 2) $-\boldsymbol{a}$ 3) $\frac{1}{2}\boldsymbol{a}$	$\boldsymbol{c} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$ Find: 1) $5\boldsymbol{c}$ 2) $-2\boldsymbol{c}$ 3) $\frac{1}{10}\boldsymbol{c}$	$\boldsymbol{p} = \begin{pmatrix} -8 \\ -20 \end{pmatrix}$ Find: 1) $0.5\boldsymbol{p}$ 2) $-3\boldsymbol{p}$ 3) $-\frac{1}{4}\boldsymbol{p}$

Q15. John writes: $2\begin{pmatrix}-3\\7\end{pmatrix}=\begin{pmatrix}-6\\7\end{pmatrix}$

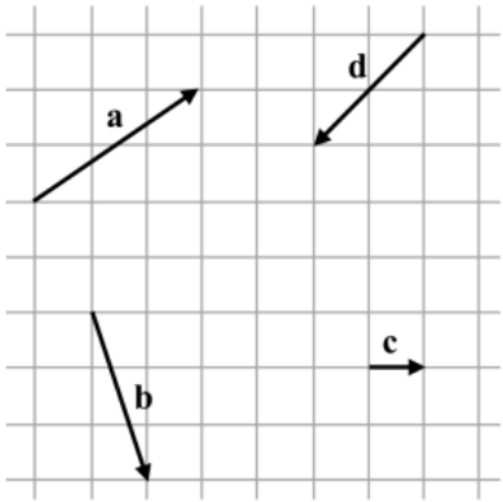
He is incorrect.
Explain why & give the correct answer.

Q16. Given that: $\boldsymbol{a}=\begin{pmatrix}2\\11\end{pmatrix}$ $\mathbf{b}=\begin{pmatrix}-8\\3\end{pmatrix}$ and $\boldsymbol{c}=\begin{pmatrix}-4\\-6\end{pmatrix}$

Write the following as column vectors.

a) $3\boldsymbol{a}$	b) $\frac{1}{2}\boldsymbol{c}$	c) $-\boldsymbol{a}$	d) $-\boldsymbol{b}$
e) $-\boldsymbol{c}$	f) $-2\boldsymbol{a}$	g) $-\frac{1}{2}\boldsymbol{c}$	h) $-5\mathbf{b}$

Q17. Work out the following, giving your answers as column vectors:



a) $2\boldsymbol{c}$

b) $2\boldsymbol{a}$

c) $3\boldsymbol{b}$

d) $\frac{1}{2}d$

e) $-\boldsymbol{c}$

f) $-\boldsymbol{b}$

g) $-5\boldsymbol{a}$

h) $-\frac{1}{2}\boldsymbol{d}$

Skills Check

$\boldsymbol{p}=\begin{pmatrix}-4\\8\end{pmatrix}$

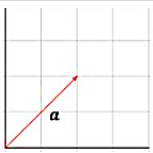
$3\boldsymbol{p}=?$

A $\begin{pmatrix}-1\\11\end{pmatrix}$ B $\begin{pmatrix}-12\\8\end{pmatrix}$ C $\begin{pmatrix}-12\\24\end{pmatrix}$ D $\begin{pmatrix}12\\24\end{pmatrix}$

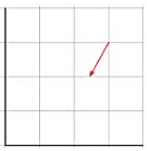
$-5\begin{pmatrix}3\\-4\end{pmatrix}=$

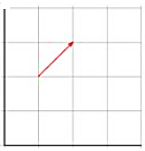
A $\begin{pmatrix}-15\\-20\end{pmatrix}$ B $\begin{pmatrix}-15\\20\end{pmatrix}$ C $\begin{pmatrix}-15\\-4\end{pmatrix}$ D $\begin{pmatrix}-2\\-9\end{pmatrix}$

The grid shows vector $\boldsymbol{a}$.
Which of the following is $\frac{1}{2}\boldsymbol{a}$?



A

B 

C 

D

$\boldsymbol{p}=\begin{pmatrix}3\\-6\end{pmatrix}$

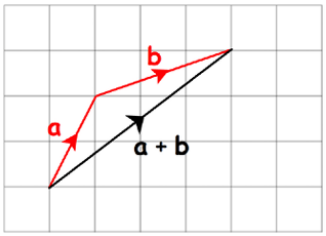
$-\frac{2}{3}\boldsymbol{p}=?$

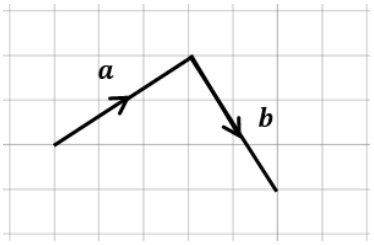
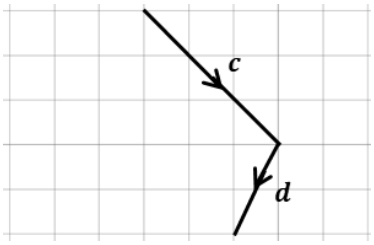
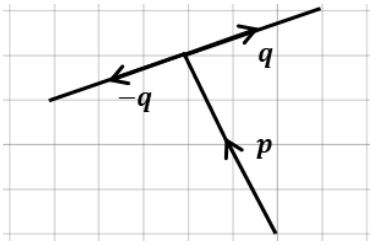
A $\begin{pmatrix}1\\-9\end{pmatrix}$ B $\begin{pmatrix}2\\-4\end{pmatrix}$ C $\begin{pmatrix}-2\\4\end{pmatrix}$ D $\begin{pmatrix}-1\\2\end{pmatrix}$

Subtopic 1C: Adding and Subtracting Vectors

1. We can add vectors together to create a combined movement, like
2. joining steps to form a path.
3. This new path is called the **resultant vector** – it replaces all the joined
4. vectors with a single one.

5. To add vectors, we combine their horizontal and vertical components
6. separately.

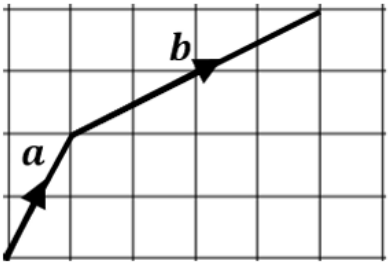


Steps	*Example 8. I do...*	*We do...*	*You do...*
1. Find the start point of the first vector. 2. Follow the given pathway to the end point of the second vector. 3. Join these 2 points and draw an arrow in the correct direction.	Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown below.  Draw the vector $\boldsymbol{a}+\boldsymbol{b}$	Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown below.  Draw the vector $\boldsymbol{c}+\boldsymbol{d}$	Vectors $\boldsymbol{p}$ & $\boldsymbol{q}$ are shown below.  Draw the vector $\boldsymbol{p}+\boldsymbol{q}$
	Write this addition in column vector notation.	Write this addition in column vector notation.	Write this addition in column vector notation.

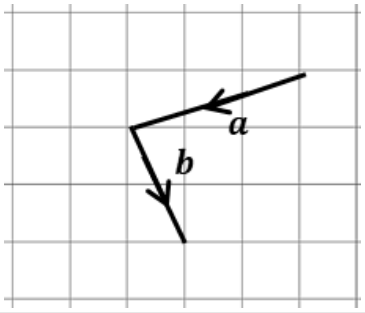
Steps	*Example 9. I do...*	*We do...*	*You do...*
1. Write the addition using column vectors. 2. Add the horizontal parts. 3. Add the vertical parts.	$\boldsymbol{a}=\begin{pmatrix}6\\1\end{pmatrix}$, $\boldsymbol{b}=\begin{pmatrix}4\\3\end{pmatrix}$ Calculate $\boldsymbol{a}+\boldsymbol{b}$	$\boldsymbol{x}=\begin{pmatrix}-3\\5\end{pmatrix}, \boldsymbol{y}=\begin{pmatrix}2\\1\end{pmatrix}$ Calculate $\boldsymbol{x}+\boldsymbol{y}$	$\boldsymbol{m}=\begin{pmatrix}5\\-1\end{pmatrix}, \boldsymbol{n}=\begin{pmatrix}-7\\-3\end{pmatrix}$ Calculate $\boldsymbol{m}+\boldsymbol{n}$

Independent Practice

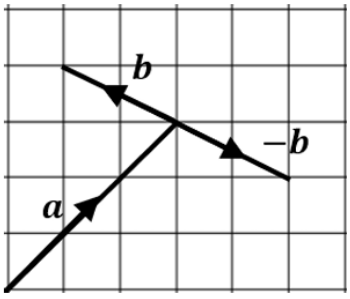
Q18. Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown. Draw vector $\boldsymbol{a}+\boldsymbol{b}$.



Q19. Vectors $\boldsymbol{a}$, & $\boldsymbol{b}$ are shown. Draw $\boldsymbol{b}+\boldsymbol{a}$.

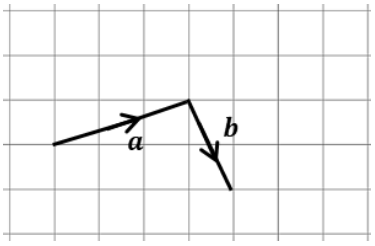
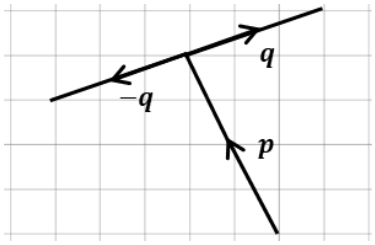
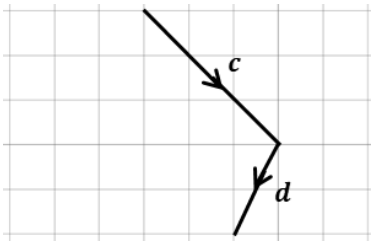


Q20. Vectors $\boldsymbol{a}$, $\boldsymbol{b}$ & $-\boldsymbol{b}$ are shown. Draw $\mathbf{a}+\boldsymbol{b}$.



Q21.	Q22.	Q23.
$\boldsymbol{x} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \quad \boldsymbol{y} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$	$\boldsymbol{a} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$	$\boldsymbol{m} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \quad \boldsymbol{n} = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$
Calculate $\boldsymbol{x} + \boldsymbol{y}$	Calculate $\boldsymbol{a} + \boldsymbol{b}$	Calculate $\boldsymbol{m} + \boldsymbol{n}$

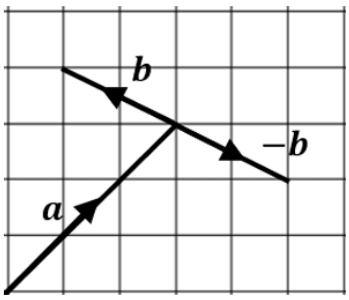
We treat **subtraction** of a vector as adding the vector in the **reverse direction**.

Steps	*Example 10. I do...*	*We do...*	*You do...*
1. Draw in the negative (reverse) vector where required. 2. Follow the pathway as before. 3. Join the start and end points, with an arrow in the correct direction.	Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown below.  Draw the vector $\boldsymbol{a} - \boldsymbol{b}$	Vectors $\boldsymbol{p}$, $\boldsymbol{q}$ & $-\boldsymbol{q}$ are shown below.  Draw the vector $\boldsymbol{p} - \boldsymbol{q}$	Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown below.  Draw the vector $\boldsymbol{d} - \boldsymbol{c}$
	Write this in column vector notation.	Write this in column vector notation.	Write this in column vector notation.

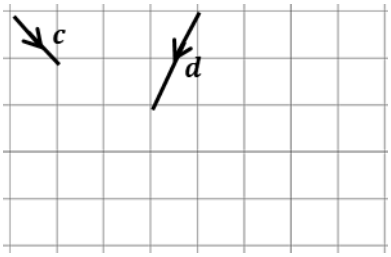
Steps	*Example 11. I do...*	*We do...*	*You do...*
1. Write the addition using column vectors. 2. Subtract the horizontal parts. 3. Subtract the vertical parts.	$\boldsymbol{a} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ Calculate $\boldsymbol{a} - \boldsymbol{b}$	$\boldsymbol{x} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}, \boldsymbol{y} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ Calculate $\boldsymbol{y} - \boldsymbol{x}$	$\boldsymbol{m} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \boldsymbol{n} = \begin{pmatrix} -7 \\ -3 \end{pmatrix}$ Calculate $\boldsymbol{m} - \boldsymbol{n}$

Independent Practice

Q24. Vectors $\boldsymbol{a}$, $\boldsymbol{b}$ & $-\boldsymbol{b}$ are shown.
Draw $\mathbf{a} - \boldsymbol{b}$

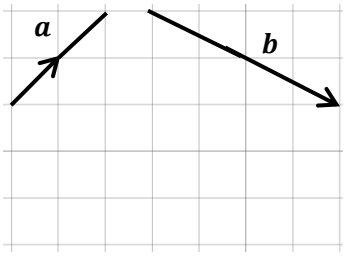
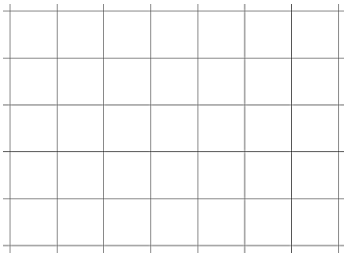
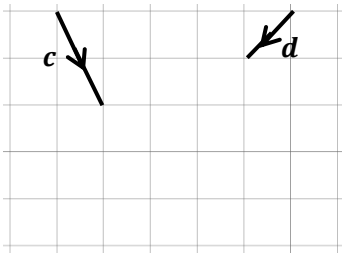
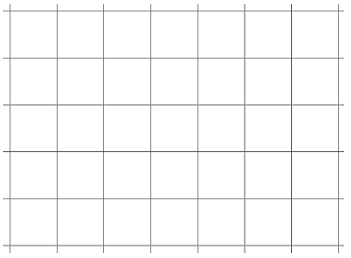
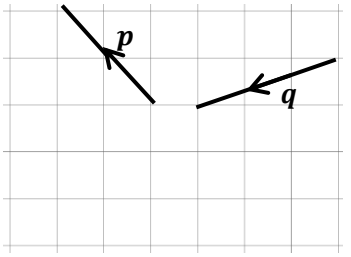
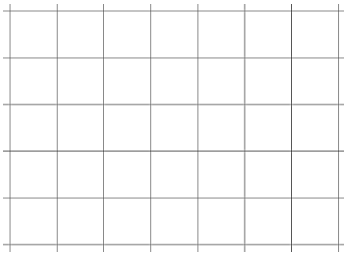


Q25. Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown. Draw $\boldsymbol{c} - \boldsymbol{d}$.



Q26.	Q27.	Q28.
$\boldsymbol{x} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \quad \boldsymbol{y} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ Calculate $\boldsymbol{x} - \boldsymbol{y}$	$\boldsymbol{a} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ Calculate $\boldsymbol{a} - \boldsymbol{b}$	$\boldsymbol{m} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \quad \boldsymbol{n} = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$ Calculate $\boldsymbol{m} - \boldsymbol{n}$

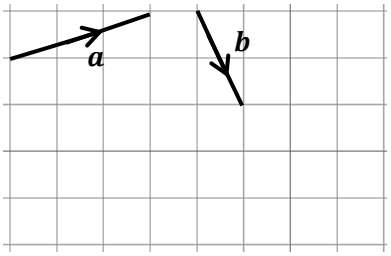
We can also add and subtract **multiples of vectors** to give resultant vectors.

Steps	*Example 12. I do...*	*We do...*	*You do...*
1. Write the column form for the vectors shown. 2. Calculate any multiples in column notation. 3. Draw the relevant single vectors to create a pathway.	Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown below.  Draw the vector $2\boldsymbol{a} + \boldsymbol{b}$ 	Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown below.  Draw the vector $\boldsymbol{c} + 3\boldsymbol{d}$ 	Vectors $\boldsymbol{p}$ & $\boldsymbol{q}$ are shown below.  Draw the vector $\frac{1}{2}\boldsymbol{p} + \boldsymbol{q}$ 
	Write this addition in column vector notation.	Write this addition in column vector notation.	Write this addition in column vector notation.

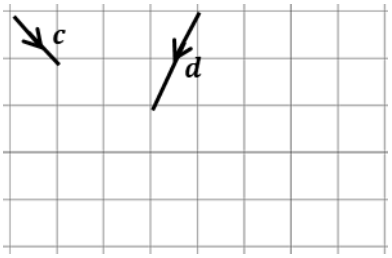
Steps	*Example 13. I do...*	*We do...*	*You do...*
1. Calculate any multiples required in column notation. 2. Complete the calculation using the horizontal and vertical parts separately.	$\boldsymbol{a} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ Calculate $2\boldsymbol{a} + \boldsymbol{b}$	$\boldsymbol{x} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}, \boldsymbol{y} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ Calculate $3\boldsymbol{y} + 2\boldsymbol{x}$	$\boldsymbol{m} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \boldsymbol{n} = \begin{pmatrix} -7 \\ -3 \end{pmatrix}$ Calculate $4\boldsymbol{m} + 3\boldsymbol{n}$

Independent Practice

Q29. Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown.
Draw $\mathbf{a} + 2\boldsymbol{b}$



Q30. Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown. Draw $2\boldsymbol{c} + \boldsymbol{d}$.



Q31.

$$\boldsymbol{x} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \qquad \boldsymbol{y} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

Calculate $\boldsymbol{x} + 4\boldsymbol{y}$

Q32.

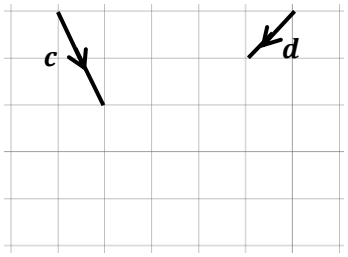
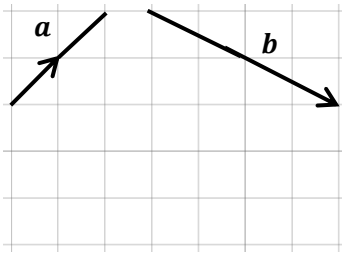
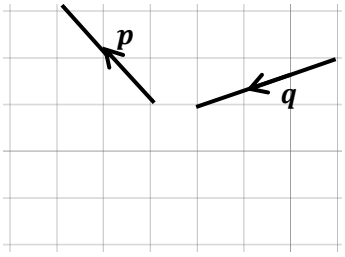
$$\boldsymbol{a} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

Calculate $\mathbf{3}\boldsymbol{a} + \mathbf{2}\boldsymbol{b}$

Q33.

$$\boldsymbol{a} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$$

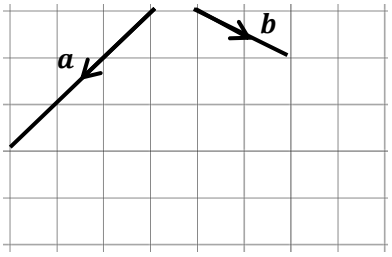
Calculate $\mathbf{5}\boldsymbol{a} + \frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{b}$

Steps	*Example 14. I do...*	*We do...*	*You do...*
1. Write the column form for the vectors shown. 2. Calculate any multiples in column notation. (Remember that subtraction can be written as the addition of the reverse vector.) 3. Draw the relevant single vectors to create a pathway.	Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown below.  Draw the vector $2\boldsymbol{c} - \boldsymbol{d}$	Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown below.  Draw the vector $\boldsymbol{a} - \frac{1}{2}\boldsymbol{b}$	Vectors $\boldsymbol{p}$ & $\boldsymbol{q}$ are shown below.  Draw the vector $-\mathbf{2}\boldsymbol{q} - \boldsymbol{p}$
	Write this addition in column vector notation.	Write this addition in column vector notation.	Write this addition in column vector notation.

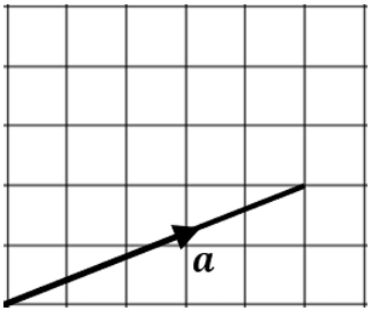
Steps	Example 15. I do...	We do...	You do...
1. Calculate any multiples required in column notation. 2. Complete the calculation using the horizontal and vertical parts separately.	$\boldsymbol{a} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$, $\boldsymbol{b} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ Calculate $2\boldsymbol{a} - \boldsymbol{b}$	$\boldsymbol{x} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}, \boldsymbol{y} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ Calculate $3\boldsymbol{y} - \frac{1}{2}\boldsymbol{x}$	$\boldsymbol{m} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \boldsymbol{n} = \begin{pmatrix} -7 \\ -3 \end{pmatrix}$ Calculate $4\boldsymbol{m} - 3\boldsymbol{n}$

Mixed Practice

Q34. Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown.
Draw vector $\frac{1}{3}\mathbf{a} + 2\boldsymbol{b}$.



Q35.
$\boldsymbol{a} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$
$\boldsymbol{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$
Draw the vector $\boldsymbol{a} - \boldsymbol{b}$ and find its column vector.



Q36. $\begin{pmatrix} 3 \\ -2 \end{pmatrix} + 2\begin{pmatrix} -7 \\ 3 \end{pmatrix} =$

Q37. $3\begin{pmatrix} 5 \\ -6 \end{pmatrix} - \begin{pmatrix} -8 \\ 2 \end{pmatrix} =$

Q38. $4\begin{pmatrix} -5 \\ 3 \end{pmatrix} - 3\begin{pmatrix} -3 \\ -8 \end{pmatrix} =$

Q39. Given that $\boldsymbol{a} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -1 \\ 6 \end{pmatrix}$ $\boldsymbol{c} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$ and $\boldsymbol{d} = \begin{pmatrix} -4 \\ -3 \end{pmatrix}$, calculate:

a) $\boldsymbol{a} + \boldsymbol{c}$	b) $\boldsymbol{b} - \boldsymbol{d}$	c) $\boldsymbol{a} + \boldsymbol{c} - \boldsymbol{b}$	d) $2\boldsymbol{a} + 0.5\boldsymbol{c}$
e) $3\boldsymbol{c} - \boldsymbol{a}$	f) $-3\boldsymbol{d} + 5\boldsymbol{b}$	g) $3(\boldsymbol{a} - \boldsymbol{b})$	h) $2\boldsymbol{a} - \boldsymbol{b} + \frac{1}{4}\boldsymbol{c} - \boldsymbol{d}$

Q40. Find the values of x and y if:

$$\begin{pmatrix}14\\-4\end{pmatrix}+\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}10\\3\end{pmatrix}$$

Q41. Find the values of p & q if:

$$\begin{pmatrix}4\\5\end{pmatrix}-2\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}-6\\7\end{pmatrix}$$

Q42. Find the values of g and h if:

$$g\begin{pmatrix}5\\4\end{pmatrix}-3\begin{pmatrix}2\\6\end{pmatrix}=\begin{pmatrix}9\\h\end{pmatrix}$$

Q43. Find the values of p & q if:

$$\begin{pmatrix}-3\\7\end{pmatrix}-p\begin{pmatrix}2\\-4\end{pmatrix}=\begin{pmatrix}-13\\q\end{pmatrix}$$

Skills Check

$$\begin{pmatrix}\mathbf{5}\\\mathbf{2}\end{pmatrix}+\begin{pmatrix}\mathbf{6}\\\mathbf{-7}\end{pmatrix}=\ ?$$

A $\begin{pmatrix}30\\-14\end{pmatrix}$ B $\begin{pmatrix}12\\-35\end{pmatrix}$

C $\begin{pmatrix}11\\-5\end{pmatrix}$ D $\begin{pmatrix}11\\9\end{pmatrix}$

$$\begin{pmatrix}\mathbf{5}\\\mathbf{2}\end{pmatrix}-\begin{pmatrix}\mathbf{6}\\\mathbf{-7}\end{pmatrix}=\ ?$$

A $\begin{pmatrix}-1\\-5\end{pmatrix}$ B $\begin{pmatrix}-1\\5\end{pmatrix}$

C $\begin{pmatrix}11\\-5\end{pmatrix}$ D $\begin{pmatrix}-1\\9\end{pmatrix}$

$$\begin{pmatrix}\mathbf{5}\\\mathbf{2}\end{pmatrix}+\ ?\ =\begin{pmatrix}\mathbf{6}\\\mathbf{-7}\end{pmatrix}$$

A $\begin{pmatrix}1\\-5\end{pmatrix}$ B $\begin{pmatrix}1\\-9\end{pmatrix}$

C $\begin{pmatrix}11\\-5\end{pmatrix}$ D $\begin{pmatrix}-1\\9\end{pmatrix}$

c = $\begin{pmatrix}-2\\-5\end{pmatrix}$ **d** = $\begin{pmatrix}4\\-4\end{pmatrix}$ Work out the vector 2**c** + 3**d**

A $\begin{pmatrix}8\\2\end{pmatrix}$ B $\begin{pmatrix}8\\-22\end{pmatrix}$

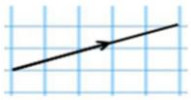
C $\begin{pmatrix}-48\\-22\end{pmatrix}$ D $\begin{pmatrix}8\\120\end{pmatrix}$

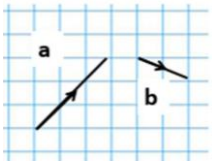
e = $\begin{pmatrix}-4\\9\end{pmatrix}$ **f** = $\begin{pmatrix}-3\\4\end{pmatrix}$ Work out the vector **e** – 2**f**

A $\begin{pmatrix}10\\-1\end{pmatrix}$ B $\begin{pmatrix}-10\\1\end{pmatrix}$

C $\begin{pmatrix}2\\1\end{pmatrix}$ D $\begin{pmatrix}2\\-1\end{pmatrix}$

The grid shows vectors $\boldsymbol{a}$ & $\boldsymbol{b}$. Which of the following is shown?

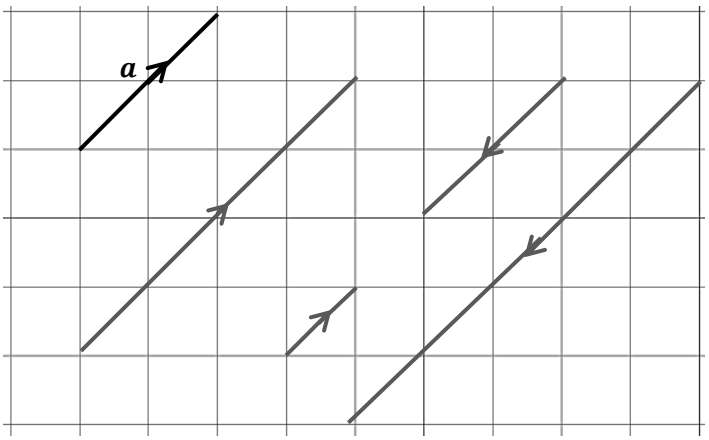




A $\boldsymbol{a}+\boldsymbol{b}$ B $\boldsymbol{a}-\boldsymbol{b}$

C $\boldsymbol{a}+2\boldsymbol{b}$ D None of these

Subtopic 2: Parallel Vectors



What is the same?

What is different?

Write these vectors in terms of $\boldsymbol{a}$

Write each vector in column notation.

Vectors are parallel if _____.

Steps	*Example 1. I do...*	*We do...*	*You do...*
1) Find a multiple, K, linking the two vectors such that $K\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}x\\y\end{pmatrix}$ 2) Write this as $\mathrm{K}\boldsymbol{a}=\boldsymbol{x}$	$\boldsymbol{a}=\begin{pmatrix}2\\-3\end{pmatrix}$ Show that the vector $\begin{pmatrix}6\\-9\end{pmatrix}$ is parallel to $\boldsymbol{a}$.	$\boldsymbol{b}=\begin{pmatrix}12\\-20\end{pmatrix}$ Show that $\begin{pmatrix}-3\\5\end{pmatrix}$ is parallel to $\boldsymbol{b}$.	$\boldsymbol{c}=\begin{pmatrix}-6\\10\end{pmatrix}$ Show that the vector $\begin{pmatrix}-15\\25\end{pmatrix}$ is parallel to $\boldsymbol{c}$.

Steps	*Example 2. I do...*	*We do...*	*You do...*
1) Find a multiple, K, linking the two vectors from the given components. 2) Write this as $\mathrm{K}\boldsymbol{a}=\boldsymbol{x}$ 3) Find the missing value	$\boldsymbol{a}=\begin{pmatrix}-7\\3\end{pmatrix}$ If $\begin{pmatrix}-21\\x\end{pmatrix}$ is parallel to $\boldsymbol{a}$, find the value of x.	$\boldsymbol{b}=\begin{pmatrix}24\\18\end{pmatrix}$ $\begin{pmatrix}-4\\y\end{pmatrix}$ is parallel to $\boldsymbol{b}$. Find the value of y.	$\boldsymbol{c}=\begin{pmatrix}-12\\21\end{pmatrix}$ $\boldsymbol{c}$ is parallel to $\begin{pmatrix}30\\z\end{pmatrix}$ Find the value of z.

Steps	Example 3. I do...	We do...	You do...
1) Complete the required column vector arithmetic 2) Find a multiple, K, linking the resultant and given vector. 3) Write this as $\mathrm{K}\boldsymbol{a} = \boldsymbol{x}$	$\boldsymbol{a} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$ Show that $\boldsymbol{a} + \boldsymbol{b}$ is parallel to the vector $\begin{pmatrix} 15 \\ 3 \end{pmatrix}$.	$\boldsymbol{a} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -6 \\ 8 \end{pmatrix}$ Show that $\boldsymbol{a} - \boldsymbol{b}$ is parallel to the vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.	$\boldsymbol{a} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$ Show that $2\boldsymbol{a} + \boldsymbol{b}$ is parallel to the vector $\begin{pmatrix} -15 \\ -9 \end{pmatrix}$

Steps	Example 4. I do...	We do...	You do...
1) Complete the required column vector arithmetic 2) Find a multiple, K, linking the resultant and given vector components. 3) Write this as $\mathrm{K}\boldsymbol{a} = \boldsymbol{x}$ 4) Find the missing value	$\boldsymbol{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$ If $\boldsymbol{a} + \boldsymbol{b}$ is parallel to $\begin{pmatrix} x \\ 12 \end{pmatrix}$, find the value of x.	$\boldsymbol{a} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -6 \\ 7 \end{pmatrix}$ If $\boldsymbol{a} - \boldsymbol{b}$ is parallel to $\begin{pmatrix} -5 \\ y \end{pmatrix}$, find the value of y.	$\boldsymbol{a} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -8 \\ -2 \end{pmatrix}$ If $2\boldsymbol{a} + \boldsymbol{b}$ is parallel to $\begin{pmatrix} z \\ -9 \end{pmatrix}$, find the value of z.

Independent Practice

Question 1. Complete the table:

Vector 1	Vector 2	Parallel?	Same direction?	Same length?
$2\boldsymbol{b}$	$5\boldsymbol{b}$			
$2\boldsymbol{a}$	$-\boldsymbol{a}$			
$3\boldsymbol{p}$		Yes	No	Yes
$\begin{pmatrix} -1 \\ 5 \end{pmatrix}$	$\begin{pmatrix} -4 \\ \boxed{} \end{pmatrix}$	Yes		
$\begin{pmatrix} 3 \\ 4 \end{pmatrix}$	$\begin{pmatrix} 9 \\ 16 \end{pmatrix}$			
$\begin{pmatrix} 15 \\ -30 \end{pmatrix}$	$\begin{pmatrix} -2 \\ 4 \end{pmatrix}$			
$\begin{pmatrix} -3 \\ -8 \end{pmatrix}$	$\begin{pmatrix} \boxed{} \\ \boxed{} \end{pmatrix}$	Yes	Yes	Yes

Question 2. Circle the parallel vectors in each set:

a) $\begin{pmatrix}20\\8\end{pmatrix}$ $\begin{pmatrix}5\\2\end{pmatrix}$ $\begin{pmatrix}-2\\-5\end{pmatrix}$ $\begin{pmatrix}-10\\4\end{pmatrix}$

b) $\begin{pmatrix}-2\\7\end{pmatrix}$ $\begin{pmatrix}4\\5\end{pmatrix}$ $\begin{pmatrix}-12\\15\end{pmatrix}$ $\begin{pmatrix}-16\\-20\end{pmatrix}$

c) $\begin{pmatrix}15\\-20\end{pmatrix}$ $\begin{pmatrix}14\\-21\end{pmatrix}$ $\begin{pmatrix}-10\\15\end{pmatrix}$ $\begin{pmatrix}4\\-9\end{pmatrix}$

Question 3.

$$\boldsymbol{b} = \begin{pmatrix}4\\3\end{pmatrix}$$

Show that $\begin{pmatrix}36\\27\end{pmatrix}$ is parallel to $\boldsymbol{b}$.

Question 4.

$$\boldsymbol{a} = \begin{pmatrix}4\\-6\end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix}-5\\0\end{pmatrix}$$

Show that $\boldsymbol{a} - 2\boldsymbol{b}$ is parallel to $\begin{pmatrix}-21\\9\end{pmatrix}$.

Question 5.

$$\boldsymbol{a} = \begin{pmatrix}x\\4\end{pmatrix}$$

$\boldsymbol{a}$ is parallel to $\begin{pmatrix}56\\32\end{pmatrix}$

Find the value of x.

Question 6.

$$\boldsymbol{p} = \begin{pmatrix}8\\-28\end{pmatrix}$$

$\boldsymbol{p}$ is parallel to $\begin{pmatrix}-4\\y\end{pmatrix}$

Find the value of y.

Question 7.

$$\boldsymbol{a} = \begin{pmatrix}0\\4\end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix}-3\\5\end{pmatrix}$$

If $3\boldsymbol{a} - 2\boldsymbol{b}$ is parallel to $\begin{pmatrix}x\\8\end{pmatrix}$, find the value of x.

Question 8.

$$\boldsymbol{a} = \begin{pmatrix}-2\\y\end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix}-7\\-6\end{pmatrix}$$

If $4\boldsymbol{a} + \boldsymbol{b}$ is parallel to $\begin{pmatrix}3\\-2\end{pmatrix}$, find the value of y.

Exam-Style Question

Given that:

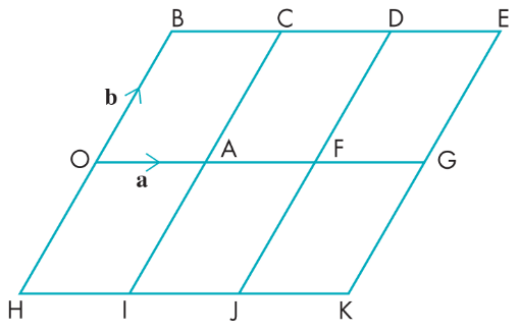
$$\boldsymbol{a} = \begin{pmatrix}2\\3\end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix}-4\\-1\end{pmatrix} \quad \boldsymbol{c} = \begin{pmatrix}5\\2\end{pmatrix}$$

a) Write $\boldsymbol{b} + \boldsymbol{c}$ as a column vector. (1)

b) Find a value of K such that $K\boldsymbol{a} + \boldsymbol{b}$ is parallel to $\boldsymbol{b} + \boldsymbol{c}$. (3)

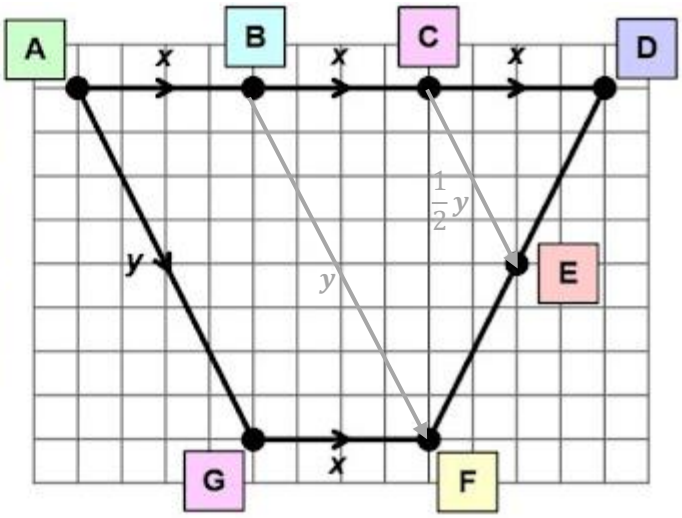
c) Give the resultant vector $K\boldsymbol{a} + \boldsymbol{b}$ as a column vector. (2)

1. Earlier on, we saw that adding and subtracting vectors can be seen as creating **a pathway**
2. **between two points**, with the resultant showing the 'direct' route between them.
3. We can use this, along with our knowledge of parallel vectors and properties of shapes to find
4. pathways between points in more complex geometrical figures.

Steps	*Example 1. I do...*	
1) Label the parallel vectors to those given in the diagram. 2) Follow a pathway between the given points. 3) Collect any like terms to simplify the final vector expression.	Write these vectors in terms of $\overrightarrow{OA} = \boldsymbol{a}$ and $\overrightarrow{OB} = \boldsymbol{b}$: *a)* $\overrightarrow{OF}$ b) $\overrightarrow{OH}$ c) $\overrightarrow{OC}$	
	We do...	***You do...***
	a) $\overrightarrow{OJ}$ b) $\overrightarrow{FH}$	*a)* $\overrightarrow{HD}$ b) $\overrightarrow{EJ}$

Vector Match: Match each of the vector expressions to a pathway below.

$2x - y$	$\frac{y}{2} + 2x$	y
$-2x$	$x - y$	$2x - \frac{1}{2}y$
$\frac{1}{2}y$	$-2x + y$	$3x - y$



$\overrightarrow{DF}$ =

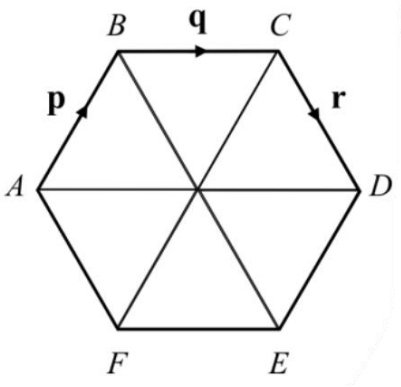
$\overrightarrow{BF}$ = $\overrightarrow{FC}$ = $\overrightarrow{GC}$ = $\overrightarrow{GD}$ =

$\overrightarrow{CE}$ = $\overrightarrow{GE}$ = $\overrightarrow{DB}$ = $\overrightarrow{AE}$ =

Question 1.

ABCDEF is a regular hexagon.

Find the vectors below in terms of $\boldsymbol{p}, \boldsymbol{q}$ and $\boldsymbol{r}$



a) $\overrightarrow{AB}$

b) $\overrightarrow{BA}$

c) $\overrightarrow{BC}$

d) $\overrightarrow{AF}$

e) $\overrightarrow{FE}$

f) $\overrightarrow{EF}$

g) $\overrightarrow{DE}$

h) $\overrightarrow{AC}$

i) $\overrightarrow{AD}$

j) $\overrightarrow{CE}$

k) $\overrightarrow{BE}$

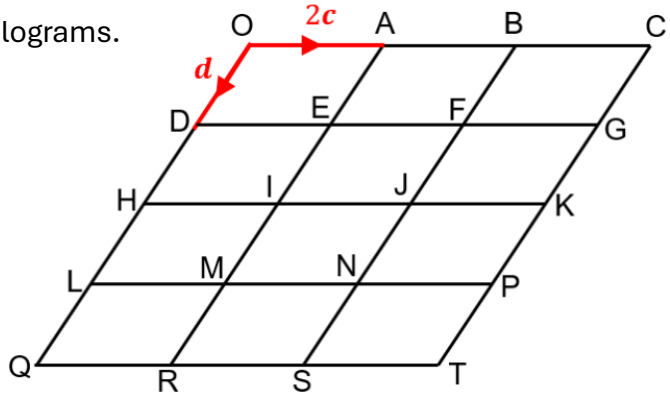
l) $\overrightarrow{DF}$

Question 2.

The diagram is made up of twelve congruent parallelograms.

$\overrightarrow{OA} = 2\boldsymbol{c}$ and $\overrightarrow{OD} = \boldsymbol{d}$.

Define the following vectors in terms of $\boldsymbol{c}$ and $\boldsymbol{d}$.



(a) $\overrightarrow{EG}$

(b) $\overrightarrow{BS}$

(c) $\overrightarrow{KJ}$

(d) $\overrightarrow{RE}$

(e) $\overrightarrow{OF}$

(f) $\overrightarrow{JT}$

(g) $\overrightarrow{MJ}$

(h) $\overrightarrow{QF}$

(i) $\overrightarrow{IL}$

(j) $\overrightarrow{HB}$

(k) $\overrightarrow{TE}$

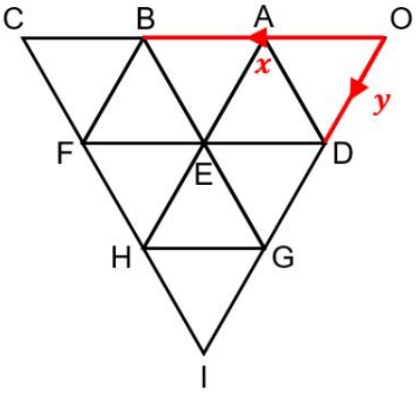
(l) $\overrightarrow{KD}$

Question 3.

The diagram is made up of nine congruent equilateral triangles.

$\overrightarrow{OB} = \boldsymbol{x}$ and $\overrightarrow{OD} = \boldsymbol{y}$.

Define the following vectors in terms of $\boldsymbol{x}$ and $\boldsymbol{y}$.



(a) $\overrightarrow{FD}$

(b) $\overrightarrow{HA}$

(c) $\overrightarrow{GH}$

(d) $\overrightarrow{OC}$

(e) $\overrightarrow{DB}$

(f) $\overrightarrow{OE}$

(g) $\overrightarrow{FO}$

(h) $\overrightarrow{CE}$

(i) $\overrightarrow{HI}$

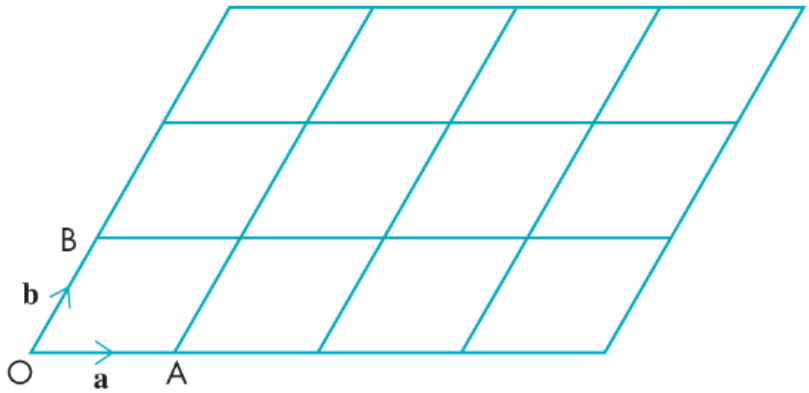
(j) $\overrightarrow{BG}$

(k) $\overrightarrow{HD}$

(l) $\overrightarrow{IF}$

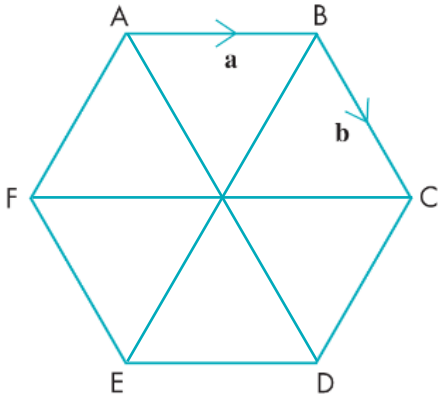
Question 4. Use the information below to mark points C-K on the grid.

a. $\overrightarrow{OC} = 2a + 3b$
b. $\overrightarrow{OD} = 2a + b$
c. $\overrightarrow{OE} = a + 2b$
d. $\overrightarrow{OF} = 3a + 3b$
e. $\overrightarrow{OG} = a + b$
f. $\overrightarrow{OH} = 2a + 2b$
g. $\overrightarrow{OI} = 3a + 2b$
h. $\overrightarrow{OJ} = 3b$
i. $\overrightarrow{OK} = 4a + b$



Question 5.

ABCDEF is a regular hexagon. $\overrightarrow{AB}$ is represented by the vector **a** and $\overrightarrow{BC}$ by the vector **b**.



a. By means of a diagram, or otherwise, show that $\overrightarrow{CD} = \mathbf{b} - \mathbf{a}$.

b. Express these vectors in terms of **a** and **b**.

i. $\overrightarrow{DE}$ ii. $\overrightarrow{EF}$ iii. $\overrightarrow{FA}$

c. Work out the resultant vector of adding:

$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} + \overrightarrow{EF} + \overrightarrow{FA}$

Explain your answer.

Q1.

$$\boldsymbol{a} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \text{ and } \boldsymbol{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

(a) Write down as a column vector

(i) $\boldsymbol{a} + \boldsymbol{b}$

.....

(1)

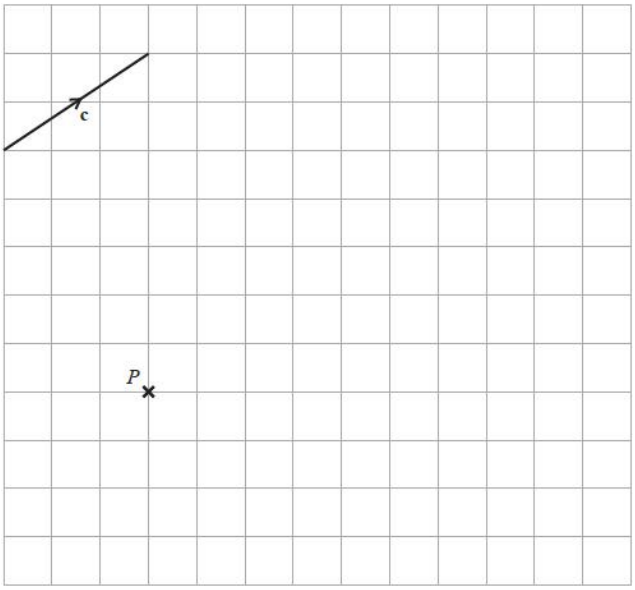
(ii) $2\boldsymbol{a} + 3\boldsymbol{b}$

.....

(2)

The vector $\boldsymbol{c}$ is drawn on the grid.

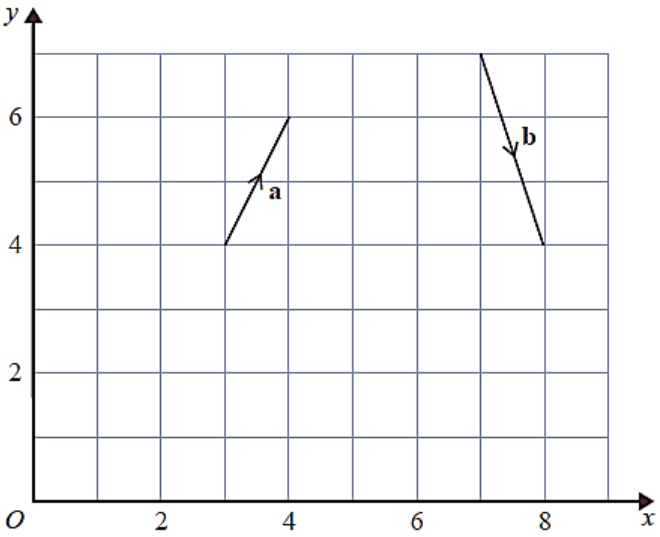
(b) From the point P, draw the vector $3\boldsymbol{c}$



(1)

(Total for question = 4 marks)

Q2. The vectors $\boldsymbol{a}$ and the vector $\boldsymbol{b}$ are shown on the grid.



a) On the grid, draw and label vector $-2\boldsymbol{a}$

(1 mark)

(b) Work out $\boldsymbol{a} + 2\boldsymbol{b}$ as a column vector.

(2 marks)

Q3. $\boldsymbol{a}$ and $\boldsymbol{b}$ are vectors such that:

$$\boldsymbol{a} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \text{ and } 3\boldsymbol{a} - 2\boldsymbol{b} = \begin{pmatrix} 8 \\ -17 \end{pmatrix}$$

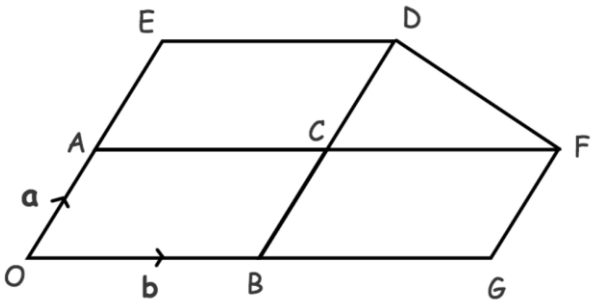
Find $\boldsymbol{b}$ as a column vector.

(3 marks)

Q4.
In the diagram OBDE and OAFG are parallelograms.

B is the midpoint of OG.
A is the midpoint of OE.

$\overrightarrow{OA} = \boldsymbol{a}$ and $\overrightarrow{OB} = \boldsymbol{b}$



a) Express in terms of a and b the following vectors, in their simplest form:

i. $\overrightarrow{OC}$

(1)

ii. $\overrightarrow{BA}$

(1)

iii. $\overrightarrow{DF}$

(1)

b) Show that $\overrightarrow{EG}$ and $\overrightarrow{DF}$ are parallel.

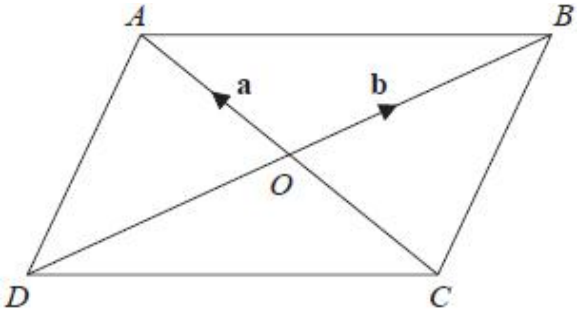
(2)

(Total marks for question is 5 marks)

Q5.
ABCD is a parallelogram.

The diagonals of the parallelogram intersect at *O*.

$\overrightarrow{OA} = \boldsymbol{a}$ and $\overrightarrow{OB} = \boldsymbol{b}$



a) Find, in terms of **b**, the vector $\overrightarrow{DB}$.

.....

(1)

(b) Find, in terms of **a** and **b**, the vector $\overrightarrow{AB}$.

.....

(1)

(c) Find, in terms of **a** and **b**, the vector $\overrightarrow{AD}$.

.....

(1)

(Total for question is 3 marks)

Subtopic 1: Right-angled Trigonometry

LEARNING OUTCOME: To understand trigonometric ratios and use them to solve problems.

Objectives:

1. To be able to label right-angled triangles using hypotenuse, opposite and adjacent.
2. To be able to select the correct trigonometric ratio from sine, cosine and tangent.
3. To be able to use the sine, cosine and tangent functions to solve problems.

Keywords:
- Hypotenuse
- Opposite
- Adjacent
- Sine
- Cosine
- Tangent

1. Trigonometry is a branch of mathematics that deals with the relationship between the **angles**
2. and **sides** of triangles. In this booklet, we'll focus on **right-angled triangles**, where one of the
3. angles is exactly 90°.
4. In a right-angled triangle, trigonometry helps us find missing side lengths or angles using **ratios**
5. connecting two sides of the triangle and a given angle. These ratios are **sine (sin), cosine (cos)**
6. **& tangent (tan).**

Subtopic 1A: Trigonometry – Building Blocks

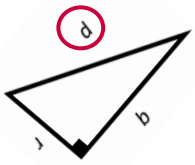
Labelling right-angled triangles

Hypotenuse

Definition
The longest side of a right-angled triangle.

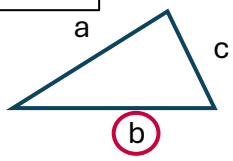
Characteristics
Only right-angled triangles have a hypotenuse.
The hypotenuse is opposite the right-angle.

Examples



This triangle is right-angled.
This line is the longest & is opposite the right-angle.
This line is a hypotenuse.

Non - Examples



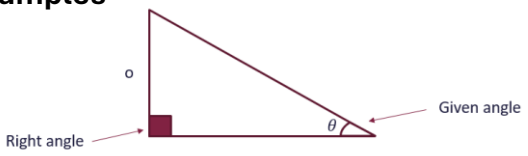
This is **not** a right-angled triangle,therefore, b is not a hypotenuse.

Opposite

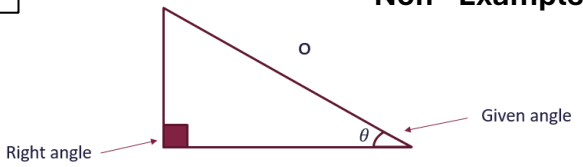
Definition
The side opposite the given angle in a right-angled triangle.

Characteristics
The opposite side must be opposite the **given angle.**

Examples



Non - Examples

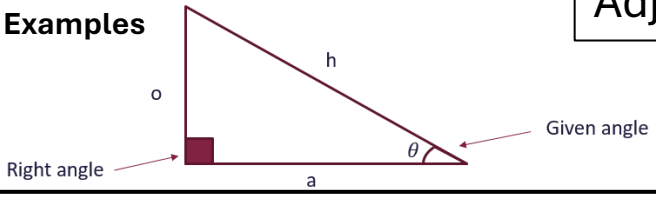


Adjacent

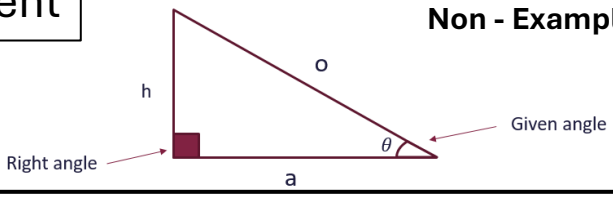
Definition
The side **next to the given** angle.

Characteristics
Adjacent is word that means **next to**.
The adjacent side will never be opposite the right-angle or the given angle.

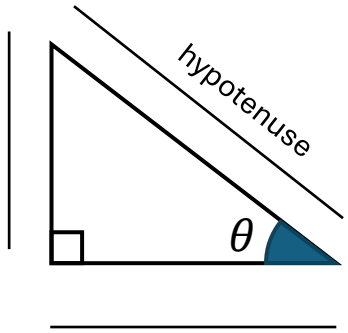
Examples

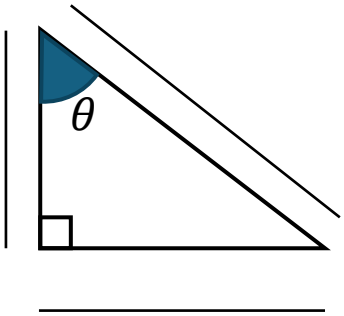


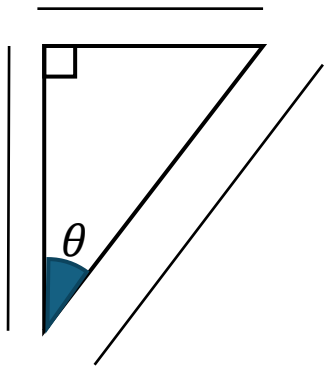
Non - Examples



Example 1. I do...	We do...	You do...
Label the triangle showing where the opposite and adjacent sides are.	Label the triangle showing where the hypotenuse, opposite & adjacent sides are.	Label the triangle showing where the hypotenuse, opposite & adjacent sides are.



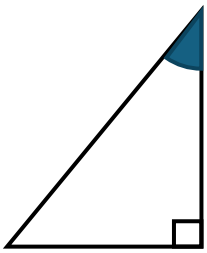




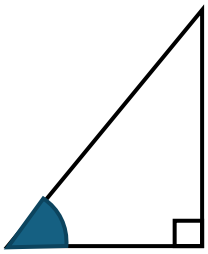
Independent Practice

Question 1. Label the hypotenuse, opposite & adjacent sides.

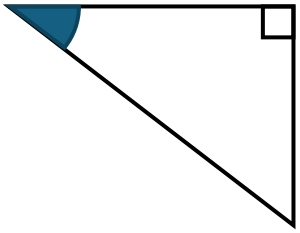
a)



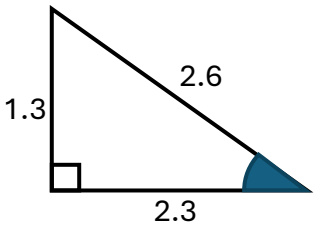
b)



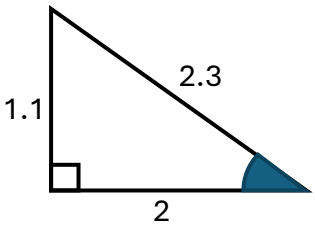
c)



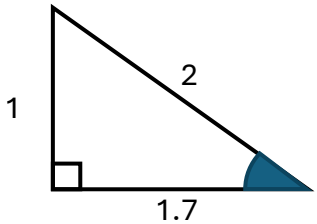
d)



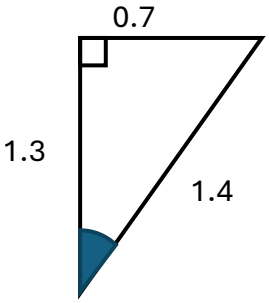
e)



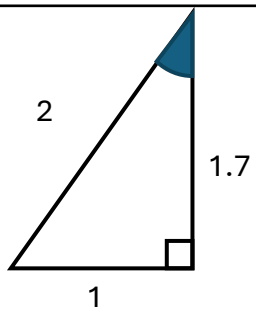
f)



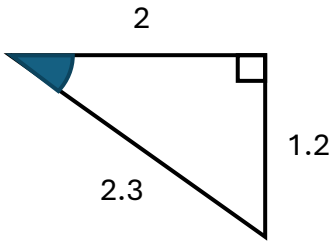
g)



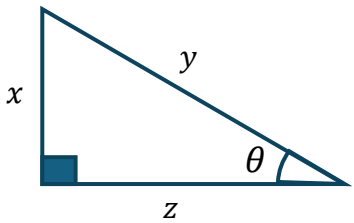
h)



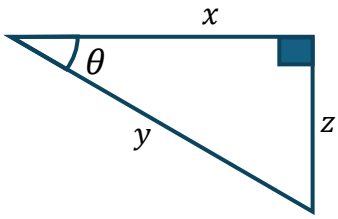
i)



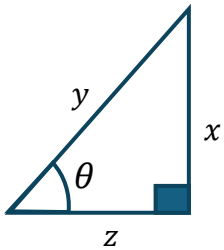
k)



l)



m)



The sine, cosine and tangent ratios for an angle θ are defined as:

$$\sin(\theta) = \frac{opposite}{hypotenuse}$$

$$\cos(\theta) = \frac{adjacent}{hypotenuse}$$

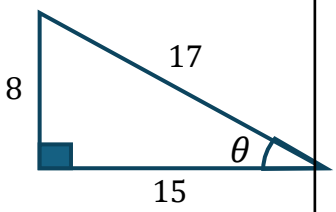
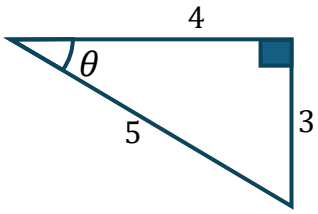
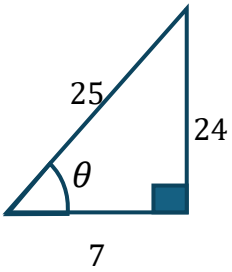
$$\tan(\theta) = \frac{opposite}{adjacent}$$

They can be abbreviated to:

$$\sin(\theta) = \frac{o}{h}$$

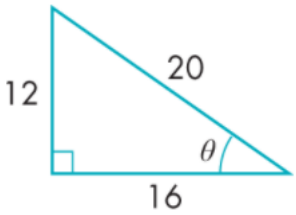
$$\cos(\theta) = \frac{a}{h}$$

$$\tan(\theta) = \frac{o}{a}$$

Steps	Complete the trigonometric ratios for each triangle:		
	Example 2. I do...	***We do...***	***You do...***
1) Write down each trigonometric ratio. 2) Substitute the known values into each ratio.			

Independent Practice **Question 2.** For each triangle define the three trigonometric ratios for angle θ.

a)

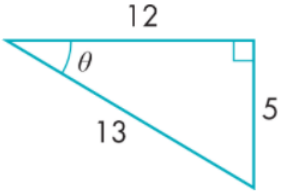


$\sin\theta =$ _____

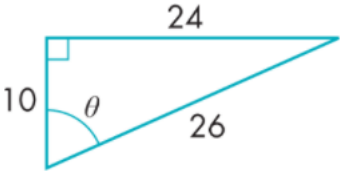
$\cos\theta =$ _____

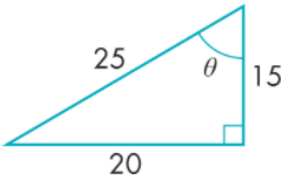
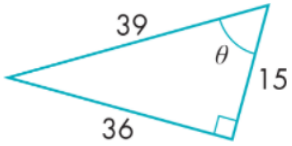
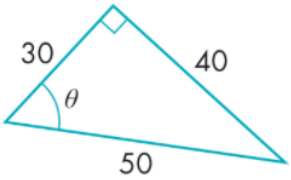
$\tan\theta =$ _____

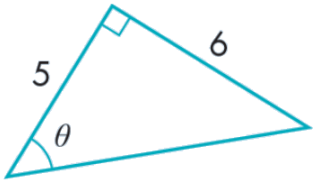
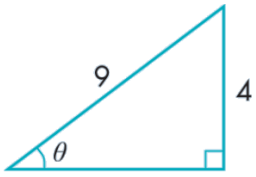
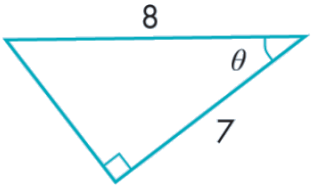
b)



c)

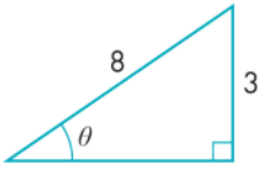
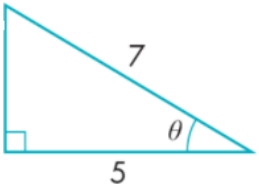
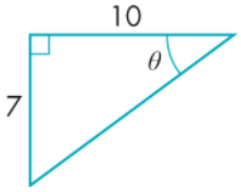
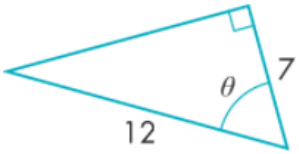
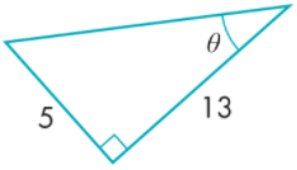
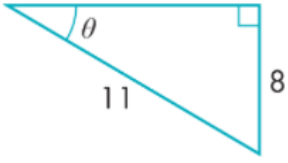


d) 	e) 	f) 

Steps	For each triangle, circle the ratio that you would use, then substitute the values in:		
	Example 3. I do...	***We do...***	***You do...***
1) Label the sides. 2) Identify which ratio these two sides belong to. 3) Substitute			

Independent Practice

Question 3. For each triangle write the appropriate trigonometric ratio:

a) 	b) 	c) 
d) 	e) 	f) 

Steps	*Example 4 I do...*	*We do...*	*You do...*
1) Type into the calculator, using the sin, cos or tan button. 2) If necessary, round to a suitable degree of accuracy.	$\sin(0)$ $\cos(20)$ $5\sin(34)$	$\cos(180)$ $\sin(80)$ $\frac{\sqrt{2}}{2}\sin(78)$	$\tan(45)$ $\cos(10)$ $1.2sin(54)$

Independent Practice

a) $\sin(10)$	b) $\cos(45)$	c) $\tan(45)$	d) $\tan(62)$
e) $\sin(85)$	f) $\tan(68)$	g) $\sin(55)$	h) $\tan(4)$
i) $5\tan(45)$	j) $4\sin(30)$	k) $8\cos(60)$	l) $6\sin(43)$
m) $9\cos(18)$	n) $15\tan(83)$	o) $14\cos(25)$	p) $\frac{\sqrt{3}}{2}\cos(45)$

Steps	*Example 5 I do...*	*We do...*	*You do...*
1) Type into the calculator, using the sin, cos or tan button. 2) If necessary, round to a suitable degree of accuracy.	$\frac{5}{\sin(34)}$	$\frac{\sin(78)}{\sqrt{2}}$	$\frac{27}{\sin(54)}$ $\frac{\tan(45)}{20}$

Independent Practice

Question 5. Evaluate the following, giving your answers to 3dp:

a) $\frac{6}{\sin(34)}$	b) $\frac{12}{\cos(83)}$	c) $\frac{\tan(16)}{4}$	d) $\frac{23}{\tan(45)}$
e) $\frac{31}{\sin(30)}$	f) $\frac{\cos(18)}{5}$	g) $\frac{48}{\tan(80)}$	h) $\frac{\sin(54)}{12}$
i) $\frac{18}{\sin(15)}$	j) $\frac{5\cos(51)}{6}$	k) $\frac{25\tan(52)}{3}$	l) $\frac{62\sin(71)}{\cos(71)}$

Steps	*Example 6 I do...*	*We do...*	*You do...*
1) Type into the calculator, using the **shift** button then the sin, cos or tan button. 2) If necessary, round to a suitable degree of accuracy. .	$\sin^{-1}(0.5)$ $\cos^{-1}(0.2)$ $\tan^{-1}\left(\frac{8}{7}\right)$	$\tan^{-1}(1)$ $\sin^{-1}(0.6)$ $\cos^{-1}\left(\frac{5}{6}\right)$	$\cos^{-1}(0.5)$ $\tan^{-1}(5)$ $\sin^{-1}\left(\frac{7}{8}\right)$

a) $\cos^{-1}(0.8)$	b) $\sin^{-1}(0.4)$	c) $\cos^{-1}(0.6)$	d) $\tan^{-1}(1.3)$
e) $\sin^{-1}\left(\frac{3}{5}\right)$	f) $\cos^{-1}\left(\frac{4}{5}\right)$	g) $\tan^{-1}\left(\frac{3}{4}\right)$	h) $\sin^{-1}\left(\frac{5}{13}\right)$
i) $\cos^{-1}\left(\frac{12}{13}\right)$	j) $\tan^{-1}(\sqrt{3})$	k) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$	l) $\cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$

Skills Check

What is the correct label for the side marked with a star?

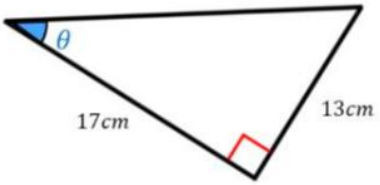
A Adjacent

B Tangent

C Hypotenuse

D Opposite

In this right-angled triangle, which trigonometric ratio would be used to find the missing angle?



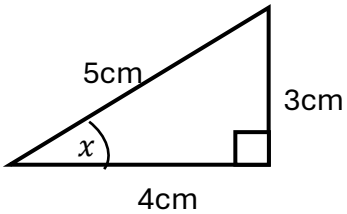
A Sin

B Cos

C Tan

D None

In this right-angled triangle, which of these is the value of $\tan(x)$?



A $\frac{3}{5}$

B $\frac{3}{4}$

C $\frac{4}{5}$

D $\frac{4}{3}$

What should replace the star?

★ $= \dfrac{Opposite}{Hypotenuse}$

A Sin

B Cos

C Tan

D None

Exam Style Questions

Q1. Use your calculator to work out $\sqrt{\dfrac{\sin(25°) + \sin(40°)}{\cos(25°) - \cos(40°)}}$

(a) Write down all the figures in your calculator display

.....

(2)

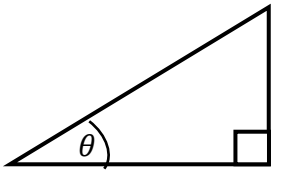
(b) Write down your answer to part (a) correct to 2 decimal places.

.....

(1)

Q2. For the right-angled triangle shown, we are given that $tan(\theta) = \frac{5}{12}$

Use this to find the value of $\sin(\theta)$



(3)

Steps	*Example 7 I do...*	*We do...*	*You do...*
1) Identify the variables that we want to isolate. & what needs to be eliminated. 2) Carry out the inverse operations to both sides.	$x \times \tan(45) = 4$	$x \times \cos(30) = 6$	$x \times \tan(60) = 5$

Independent Practice

a) $x \times \sin(45) = 4$	b) $x \times \sin(30) = 5$	c) $x \times \cos(30) = 7$	d) $x \times \tan(30) = 7$
e)	f)	g)	h)

Unknown on the Denominator

Solving a problem with a trigonometric ratio leads to an equation with a fraction. These can be tricky to solve, particularly if **the unknown is on the denominator** of the fraction.

Steps	*Example 8 I do...*	*We do...*	*You do...*
Use inverse functions to isolate the required variable.	Solve: $12 = \frac{x}{5}$	Solve: $8 = \frac{x}{2.5}$	Solve: $100 = \frac{x}{8}$
	Example 9 I do...	*We do...*	*You do...*
Where the variable is on the denominator, multiply first to make this the subject.	Solve: $12 = \frac{5}{x}$	Solve: $8 = \frac{2.5}{x}$	Solve: $100 = \frac{8}{x}$

Steps	Solve to find x. Give your answer to 2 decimal places, where necessary.		
	Example 10 I do...	***We do...***	***You do...***
1) Circle the variable that we want to isolate. 2) Identify what needs to be eliminated. 3) Carry out the inverse operation to both sides. 4) Work out the value of x using your calculator.	$\sin(30) = \frac{x}{4}$	$\cos(45) = \frac{x}{4}$	$\sin(60) = \frac{x}{3}$
	Example 11 I do...	***We do...***	***You do...***
	$sin(15) = \frac{5}{x}$	$cos(45) = \frac{2}{x}$	$tan(30) = \frac{2}{x}$

Independent Practice

Question 7. Rearrange to make x the subject, then solve to find x. Give your answer to 2 decimal places, where necessary.

a) $sin(30) = \frac{x}{2}$	b) $tan(45) = \frac{x}{2}$	c) $sin(45) = \frac{x}{2}$	d) $sin(45) = \frac{x}{4}$
e) $\frac{x}{4} = sin(45)$	f) $tan(30) = \frac{x}{4}$	g) $tan(30) = \frac{x}{2}$	h) $cos(30) = \frac{x}{2}$

Question 8. Rearrange to make x the subject, then solve to find x.
Give your answer to 2 decimal places, where necessary.

a) $cos(30) = \frac{2}{x}$	b) $cos(45) = \frac{2}{x}$	c) $sin(45) = \frac{2}{x}$	d) $sin(45) = \frac{4}{x}$
e) $sin(45) = \frac{8}{x}$	f) $sin(45) = \frac{8}{x}$	g) $sin(45) = \frac{x}{8}$	h) $cos(45) = \frac{x}{8}$
i) $cos(45) = \frac{8}{x}$	j) $\frac{8}{x} = \cos(45)$	k) $cos(30) = \frac{5}{x}$	l) $cos(60) = \frac{5}{x}$

Involving inverse sine, cosine and tangent

Sometimes the unknown variable we want is 'trapped' within the trigonometric function, so we need to use the inverse function; $\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$.

Steps	*Example 12 I do...*	*We do...*	*You do...*
1) Identify the variables that we want to isolate. & what needs to be eliminated. 2) Carry out the inverse operations to both sides.	$\sin(x) = 0.45$	$\cos(x) = \frac{3}{4}$	$\tan(x) = \frac{4}{3}$

Independent Practice

Question 9. Solve to find x, giving your answer to 2 decimal places, where necessary.

a) $\sin x = 0.707$	b) $\cos x = 0.707$	c) $\tan x = 0.707$
d) $\cos x = \frac{7}{10}$	e) $\tan x = \frac{10}{7}$	f) $\tan x = \frac{20}{14}$
g) $\sin x = 0.866$	h) $\cos x = 0.866$	i) $\cos x = \frac{1}{2}$
j) $\cos x = \frac{0.7}{0.8}$	k) $\tan x = \frac{0.7}{3.5}$	l) $\tan x = \frac{2}{3}$
m) $\cos x = \frac{\sqrt{3}}{2}$	n) $\tan x = 1$	o) $\sin x = \frac{\sqrt{2}}{2}$

Subtopic 1B: Right-angled Trigonometry: Calculating a missing length

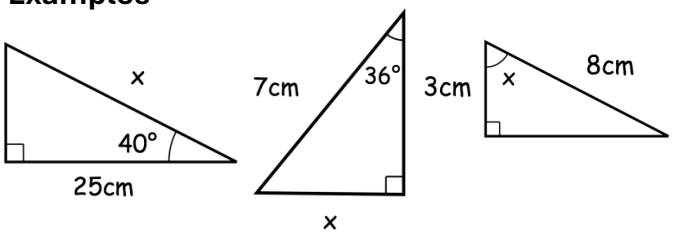
Definition
Trigonometry is a branch of mathematics that deals with the relationship between the angles and sides of triangles.

Characteristics
In a right-angled triangle, trigonometry is used when you have:

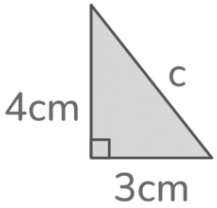
- One side, one angle and finding another side
- Two sides and finding an angle

Trigonometry

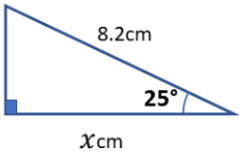
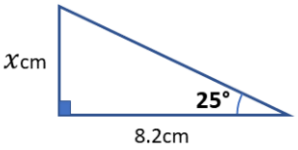
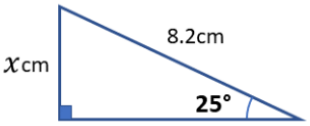
Examples



Non - Examples



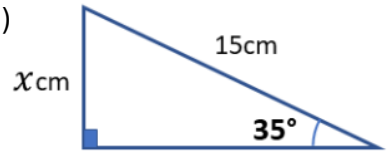
Here we have a problem involving 3 sides.
We would use Pythagoras to find c, not trigonometry!

Steps	Calculate the missing side length, x.		
	Example 1 I do...	***We do...***	***You do...***
1) Label the sides. 2) Choose the correct ratio. 3) Substitute in the known values. 4) Rearrange and solve to find x. 5) **You must show all stages of your working out.**			

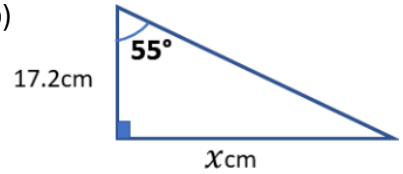
Independent Practice

Question 1. Calculate the value of x, giving your answers to 2 d.p.

a)

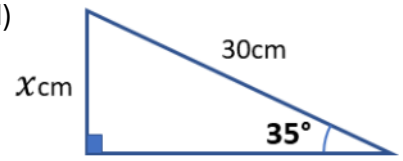


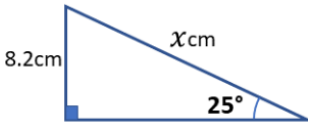
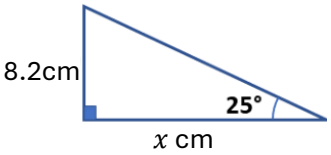
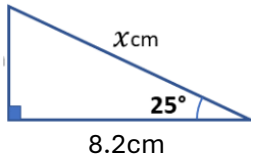
b)



c)

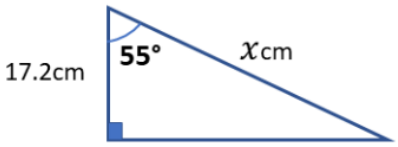
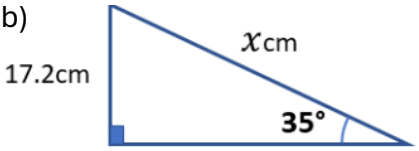
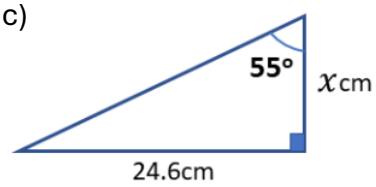
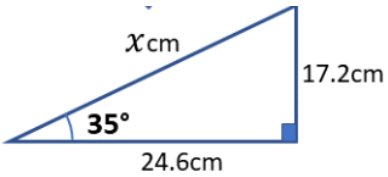
d)



Steps	Calculate the missing side length, x.		
	Example 2 I do...	***We do...***	***You do...***
1) Label the sides. 2) Choose the correct ratio. 3) Substitute in the known values. 4) Rearrange and solve to find x. 5) **You must show all stages of your working out.**			

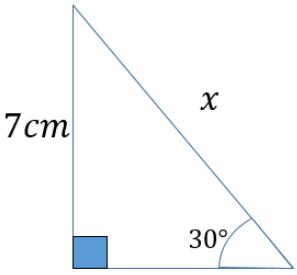
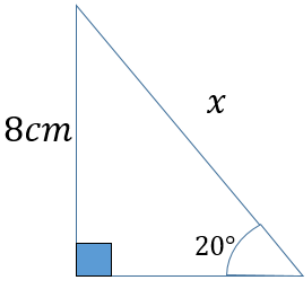
Independent Practice

2. Calculate the value of x, giving your answers to 2 decimal places.

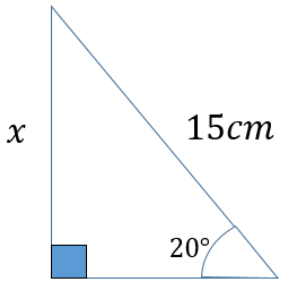
a) 	b) 
c) 	d) 

Mixed Practice

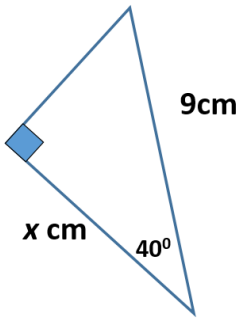
Question 2. Calculate the value of x, giving your answers to 2 decimal places.

a) 	b) 

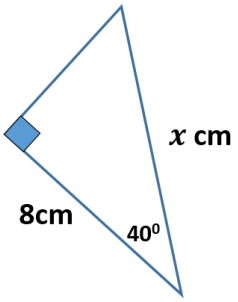
c)


x
$15cm$
20°

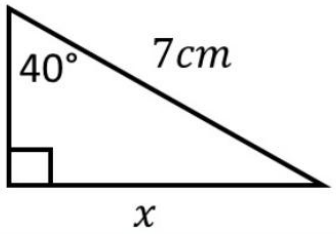
d)


9cm
x cm
40⁰

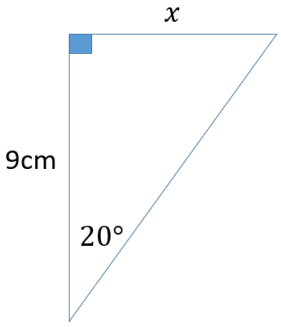
e)


x cm
8cm
40⁰

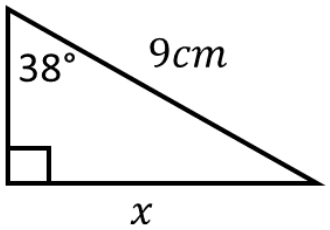
f)


40°
$7cm$
x

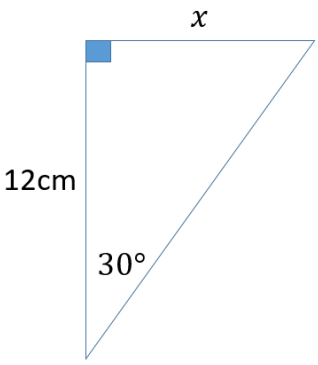
g)


x
9cm
20°

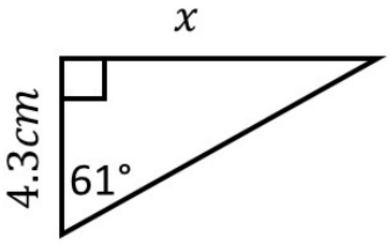
h)


38°
$9cm$
x

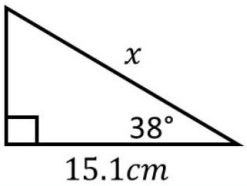
i)


x
12cm
30°

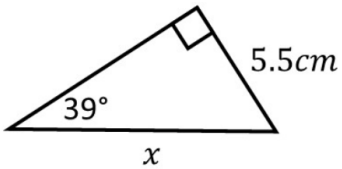
j)


x
$4.3cm$
61°

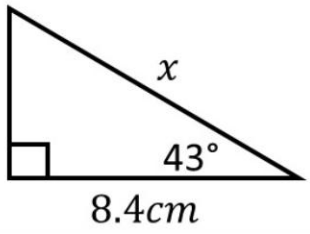
k)


x
38°
$15.1cm$

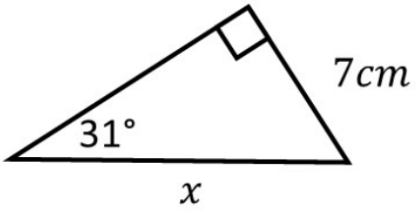
l)


$5.5cm$
39°
x

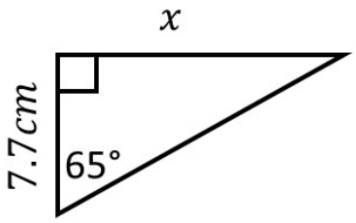
m)



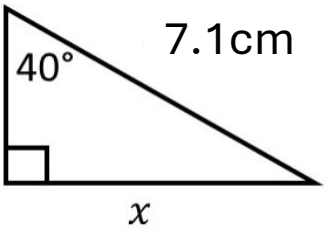
n)



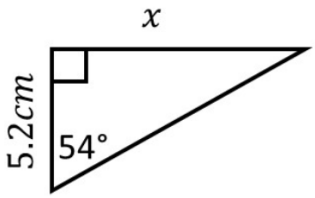
o)



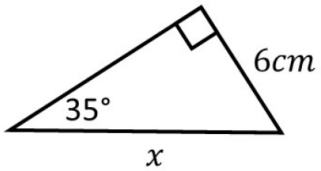
p)



q)

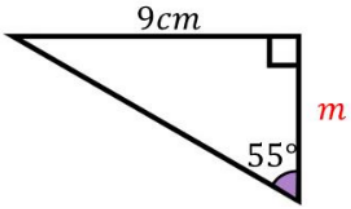


r)



Skills Check

Which is the correct ratio to work out the length of side m?



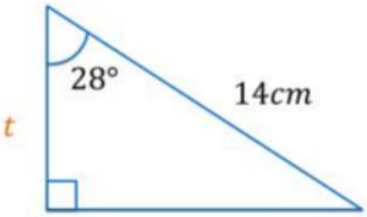
A Pythagoras

B Sin

C Cos

D Tan

Which is the correct method to work out side t?



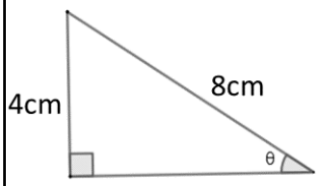
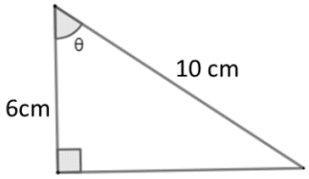
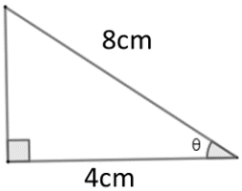
A $\frac{\cos(28)}{14}$

B $\frac{14}{\cos(28)}$

C $\sin(28) \times 14$

D $\cos(28) \times 14$

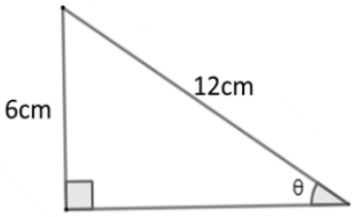
Subtopic 1C: Right-angled Trigonometry: Calculating a missing angle

Steps	Calculate the missing angle, θ		
	Example 1 I do...	***We do...***	***You do...***
1) Label the sides. 2) Choose the correct ratio. 3) Substitute in the known values. 4) Rearrange and solve to find θ. 5) **You must show all stages of your working out.**			

Independent Practice

Question 1. Calculate the value of θ, giving your answers to 2 decimal places.

a)



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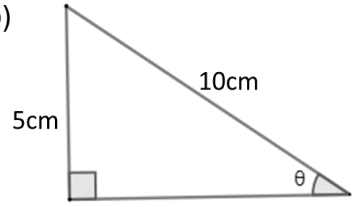

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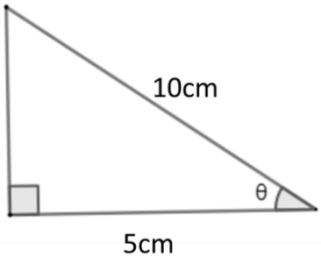

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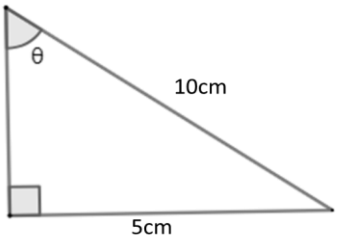

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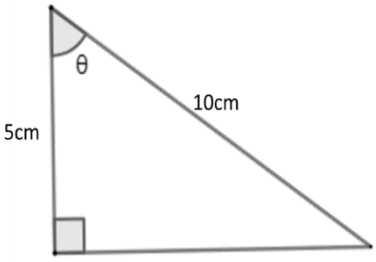

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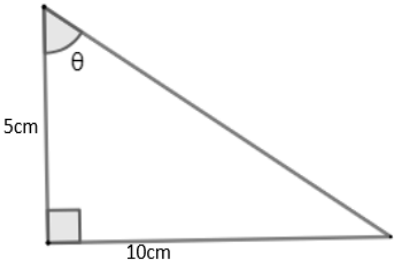

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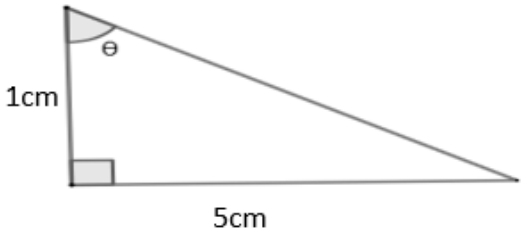
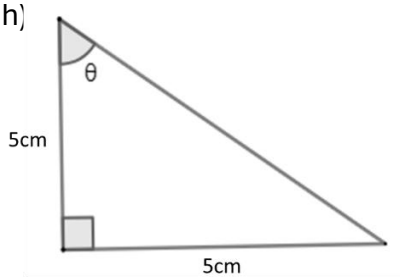
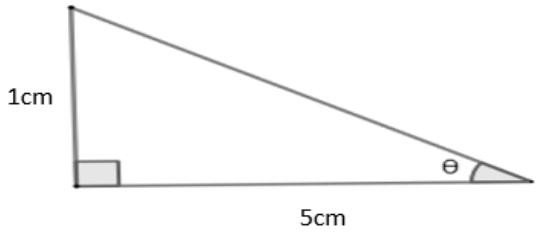

θ
1cm
5cm

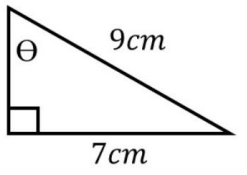
h)


θ
5cm
5cm

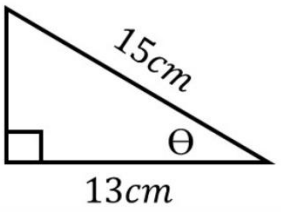
i)


1cm
θ
5cm

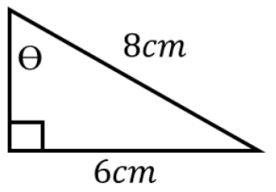
j)


θ
9cm
7cm

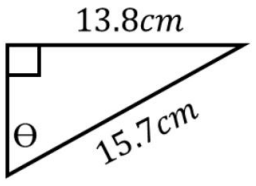
k)


15cm
θ
13cm

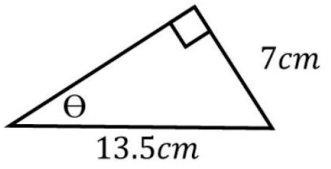
l)


θ
8cm
6cm

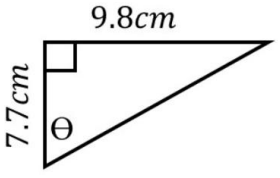
m)


13.8cm
θ
15.7cm

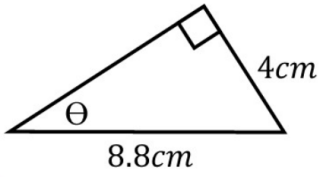
n)


7cm
θ
13.5cm

o)

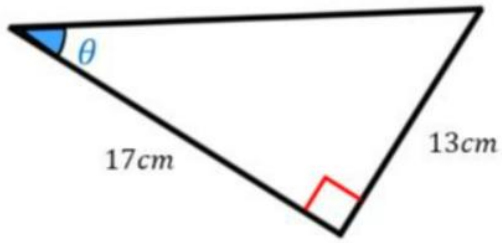

9.8cm
7.7cm
θ

p)


4cm
θ
8.8cm

Skills Check

Which ratio would you use to find the value of θ?



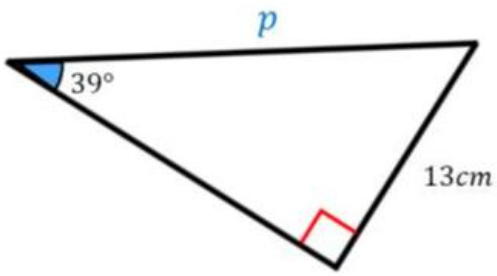
A Sin

C Tan

B Cos

D None of these

Which ratio would you use to find the value of p?



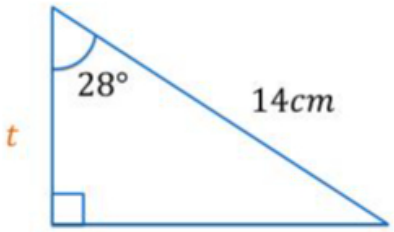
A Sin

C Tan

B Cos

D None of these

Which is the correct method to work out side t ?



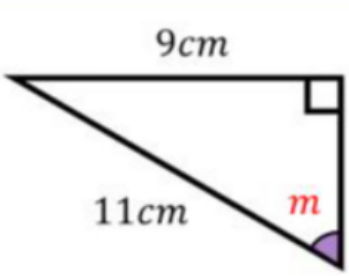
A $\frac{\cos(28)}{14}$

B $\frac{14}{\cos(28)}$

C $\sin(28) \times 14$

D $\cos(28) \times 14$

Which is the correct method to work out angle m?



A $\sin(\frac{9}{11})$

B $\cos(\frac{9}{11})$

C $\sin^{-1}(\frac{9}{11})$

D $\cos^{-1}(\frac{9}{11})$

Tom and Katie are arguing about which units could be used with the solution to $\boldsymbol{sin^{-1}(0.8)} =$

Tom says the answer could be in degrees

Katie says the answer could be in millimetres

Who is correct?

A Only Tom

B Only Katie

C Both Tom and Katie

D Neither is correct

CHALLENGE

a) Draw 3 right-angled triangles with angle θ such that $\sin(\theta) = \frac{3}{5}$	b) A right-angled triangle has hypotenuse of 5cm, an angle of 37° and the side opposite the angle of 3cm. Work out the exact value of $\sin(53)$.
c) A right-angled triangle has $\sin(x) = \frac{5}{13}$ and one side length of 60cm. What lengths could the other sides be?	d) A right-angled triangle has a side length of 60cm and an angle of 30°. What could the other sides be?

Subtopic 1D: Applications of Trigonometry

LEARNING OUTCOME: To solve problems involving right-angled trigonometry

Objectives:

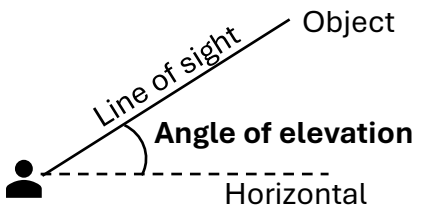
1. To be able to solve worded & multi-step right-angled trigonometry problems.
2. To be able to solve problems in non-right-angled triangles.
3. To be able to solve problems involving a mixture of Pythagoras' Theorem and right-angled trigonometry.

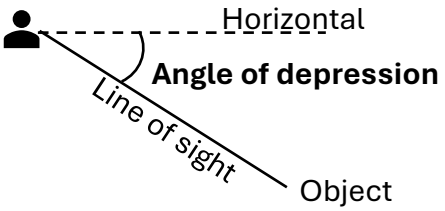
Keywords:

- Hypotenuse
- Opposite
- Adjacent
- Sine
- Cosine
- Tangent

Introduction – Read and annotate:

1. In this subtopic, you will learn how to apply right-angled trigonometry to a wide range of
2. problems. You'll practise drawing diagrams to represent word problems and tackle multi-step
3. questions.
4. You will come across 'angle of elevation' and 'angle of depression'.
5. If a person stands and looks up at an object, the **angle of elevation** is the angle between the horizontal
6. line of sight and the object.
7. If a person stands and looks down at an object, the **angle of depression** is the angle between the horizontal line of sight and the object.





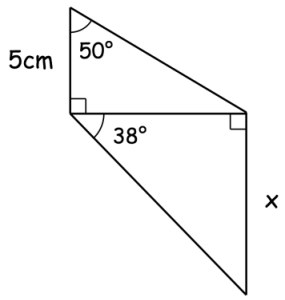
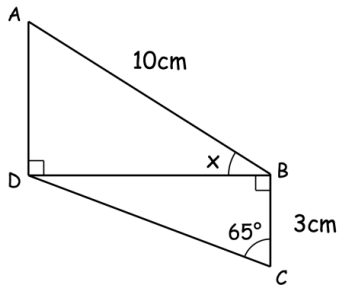
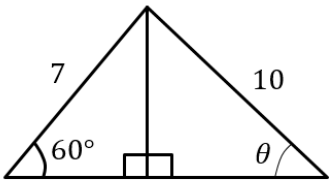
Worded Problems

Steps	*Example 1 I do...*	*We do...*	*You do...*
1) Draw a diagram to represent the situation. 2) Use trigonometry to calculate the missing value. 3) **You must show all stages of your working out.**	A plane-spotter sees a plane in the sky at an angle of elevation of $18°$. The plane is a horizontal distance of $40\ km$ from the plane-spotter. Find the vertical height of the plane in kilometres.	A 5-metre-long ladder is placed against a wall. It reaches 4.3m up the wall. What is the angle between the ladder and the ground?	A ladder makes an angle of $75°$ with the ground. The distance of the foot of the ladder to the wall is $1.45\ m$. How long is the ladder in metres?

Independent Practice

Question 1. Answer the following. **You must show all of your working out.**

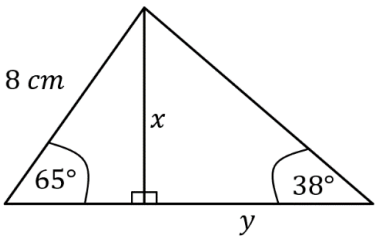
a) Malia is flying a kite on a 20 m long string. The string is at an angle of 35° to the horizontal. Malia is holding the kite 1.1 m above the ground. Find the vertical height of the kite above the ground in metres.	b) A bird sits on the ground 9 m away from the base of a Christmas tree. The angle of elevation from the bird to the top of the tree is 52°. How tall is the tree in metres?
c) A helicopter leaves Bristol and flies due east for 10 miles. Then the helicopter flies 8 miles north before landing. Calculate the bearing of the helicopter from Bristol.	d) A helicopter leaves place A and flies 40 miles due east. Then the helicopter flies 10 miles due south and arrives at place B. Work out the bearing of B from A.
e) A boat is sailing towards a port. From its current position, the port is **8 km east** and **6 km north**. To avoid shallow water, the boat must sail on a bearing between **35° and 50°** (measured clockwise from due north). Is the boat sailing within a safe range?	f) A ladder is placed against a vertical wall. The ladder is a horizontal distance of 1.5m from the wall and a vertical height of 5.8m up the wall. To be safe, it **must** be at an angle between 70° and 80° to the ground. Is the ladder safe?

Steps	*Example 2 I do...*	*We do...*	*You do...*
1) Identify and highlight the shared side. 2) Calculate the value of the shared side using trigonometry. 3) Use the value of the shared side to calculate x. 4) **You must show all stages of your working out. out.**	Calculate the value of x  	Calculate the value of x  	Calculate the value of θ 

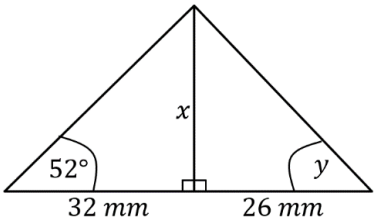
Independent Practice

Question 2. Calculate the missing lengths and angles.

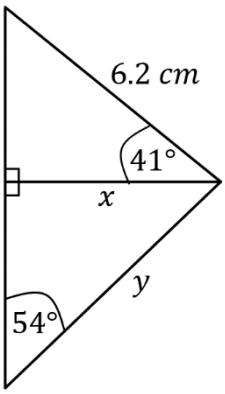
a)



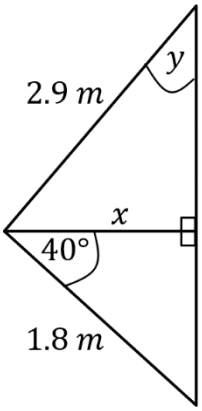
b)



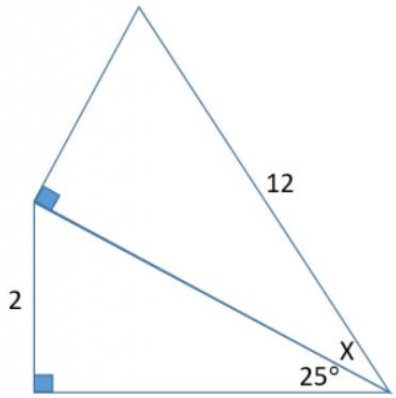
c)



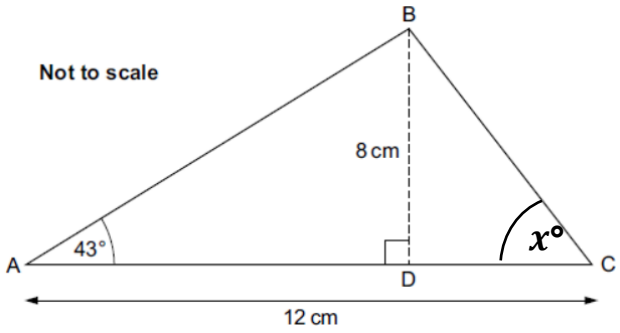
d)



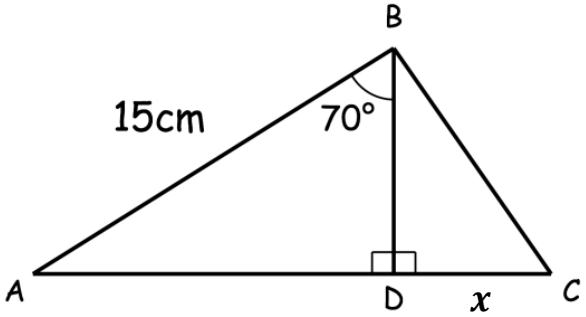
e) Calculate the value of x.



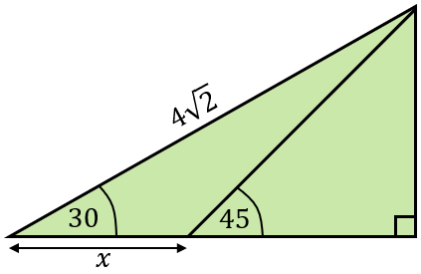
f) In the diagram, ABC is a triangle and line BD is perpendicular to AC.
Calculate angle BCA.

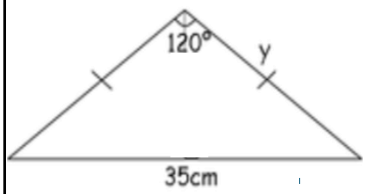
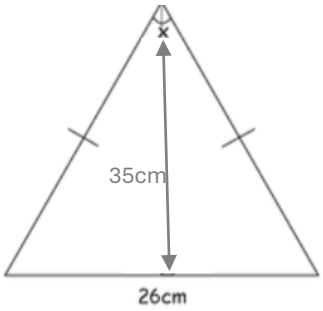
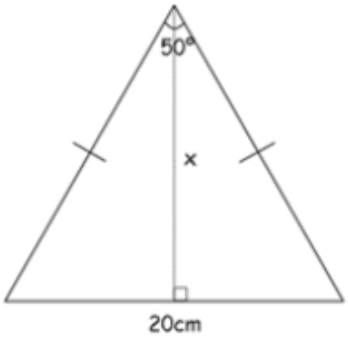


h) Shown below are right-angled triangles, ABD and BCD.
AB = 15cm, Angle ABD = 70°
Angle ABD : Angle DBC = 5 : 2
Calculate the length of CD.



i) Find the value of x.

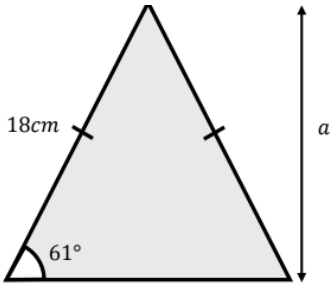


Steps	Calculate the value of the unknowns		
	Example 3 I do...	***We do...***	***You do...***
1) Draw the perpendicular line to form a right-angled triangle. 2) Draw a right-angled triangle with correct measurements. 3) Use trigonometry to find unknown. 4) Check that you have the required side or angle in full. **5) You must show all stages of your working out.**			

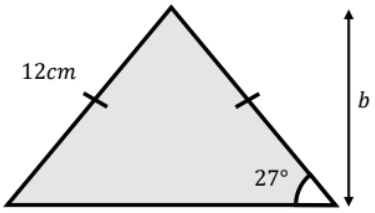
Independent Practice

Question 3. Calculate the missing lengths and angles.

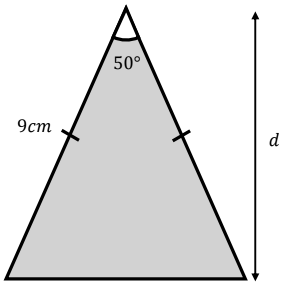
a)



b)



c)



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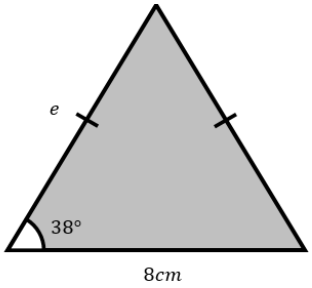
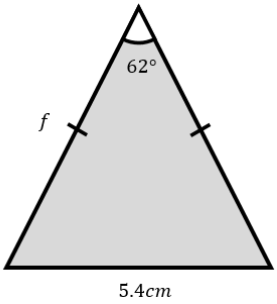
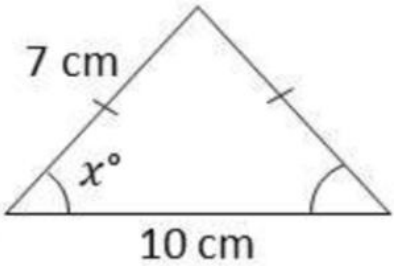


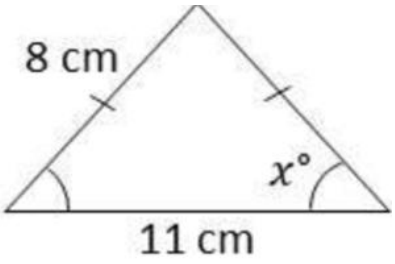
e)



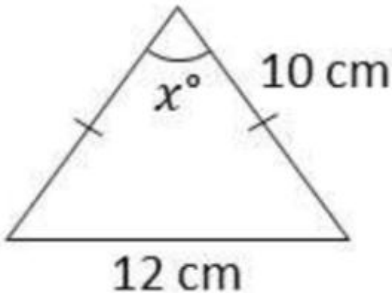
f)



g)

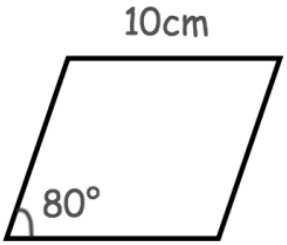


h)

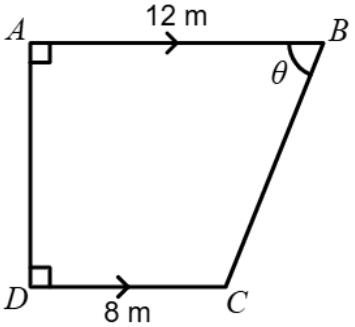


CHALLENGE: 'Hidden' Right-Angled Triangles

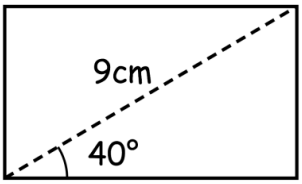
Question 4. Calculate the area of the rhombus:



Question 5. The area of the trapezium is 100cm^2. Calculate the size of angle θ

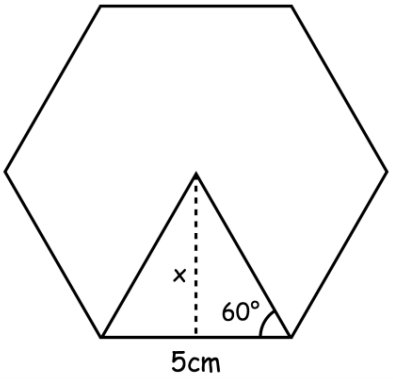


Question 6. Calculate the perimeter of the rectangle:



Question 7. A regular hexagon can be divided into 6 equilateral triangles.
The diagram below shows one of the equilateral triangles.

a) Use trigonometry to find the height, x, of the triangle.



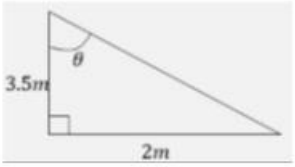
b) Calculate the area of the equilateral triangle.

c) Calculate the area of the hexagon.

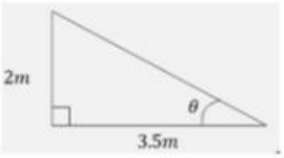
Skills Check

Erica is trying to find the angle Of elevation of the sun, using the shadow of a lamppost which is 3.5m high. If the shadow is 2m long, which diagram will help to calculate the angle of elevation of the sun?

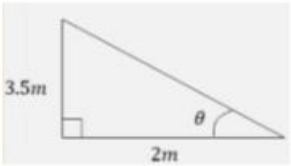
A



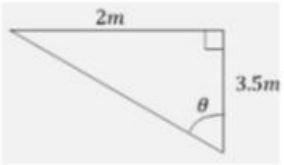
B



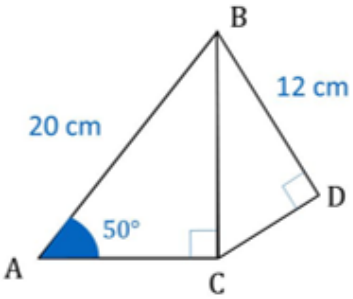
C



D



ABC and BCD are right angled triangles.
Angle ACB = Angle BDC = 90° and angle BAC = 50°
AB = 20 cm and BD = 12 cm
Find the length CD, to 1 decimal place.



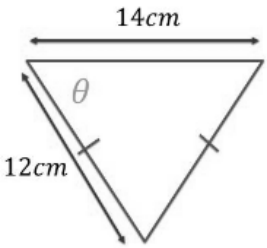
A 9.5 cm

B 4.6 cm

C 14.3 cm

D 15.3 cm

Calculate the value of θ to 1 decimal place.



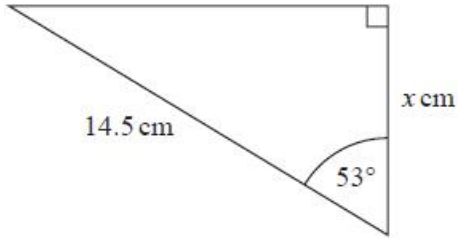
A 1.0°

B 35.7°

C 54.3°

D No possible solutions

Q1. Work out the value of x.
Give your answer correct to 3 significant figures.

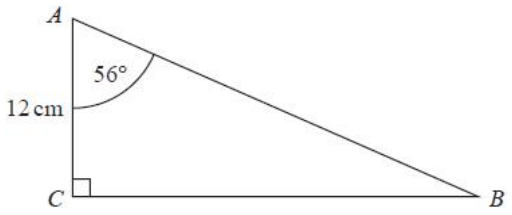


x = ...

(Total for question = 2 marks)

Q2. *ABC* is a right-angled triangle.

(a) Work out the length of *BC*.
Give your answer correct to 1 decimal place.

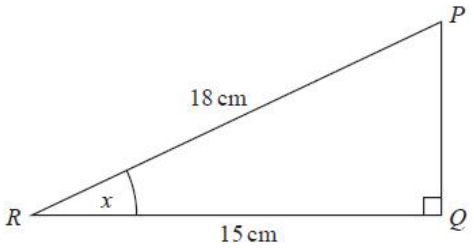


..... cm

(2)

PQR is a right-angled triangle.

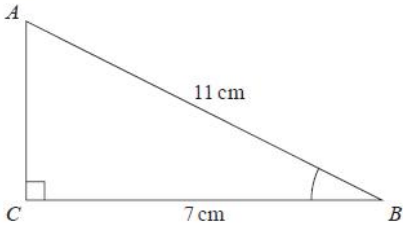
(b) Work out the size of the angle marked x.
Give your answer correct to 1 decimal place.



..... °

(2)

Q3. *ABC* is a right-angled triangle.
(a) Work out the size of angle *ABC*.
Give your answer correct to 1 decimal place.



..... °

(2)

The length of the side *AB* is reduced by 1 cm.
The length of the side *BC* is still 7 cm.
Angle *ACB* is still 90°

(b) Will the value of cos ABC increase or decrease?
You must give a reason for your answer.

.....

.....

.....

(1)

Subtopic 2: Problems with Trigonometry & Pythagoras' Theorem

Subtopic 2A: Pythagoras' Theorem - RECAP

Pythagoras' Theorem

Definition

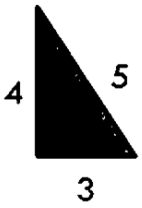
The **sum of the squares** of the two shortest sides of a right-angled triangle are equal to the square of the hypotenuse.

$$a^2 + b^2 = c^2$$

Characteristics

- You can only use Pythagoras to find missing sides in a **right-angled triangle**.
- You can use Pythagoras to find a missing hypotenuse and missing shorter sides.

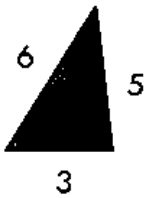
Examples



$$3^2 + 4^2 = 5^2$$

This triangle is right-angled and the conditions for Pythagoras to work are met.

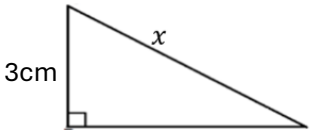
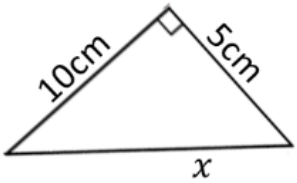
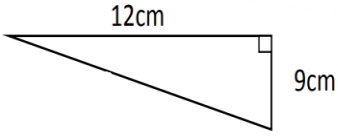
Non - Examples



This is **not** a right-angled triangle.

$$3^2 + 5^2 \neq 6^2$$

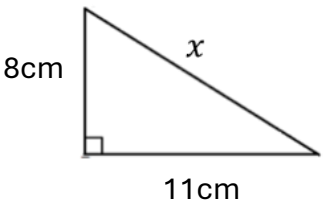
The conditions for Pythagoras are not met.

Steps	Use Pythagoras' theorem to find the missing side. Round to 1dp if necessary.		
	Example 1 I do...	***We do...***	***You do...***
1) Label the triangle (use c for the hypotenuse) 2) Substitute the values into Pythagoras' theorem $a^2 + b^2 = c^2$ 3) Rearrange to find the unknown. 4) Round your answer to the specified amount.			

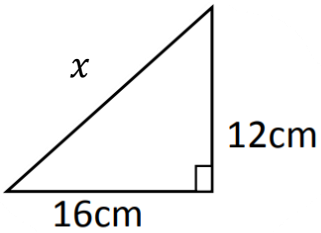
Independent Practice

Question 1. Calculate the missing lengths and angles.

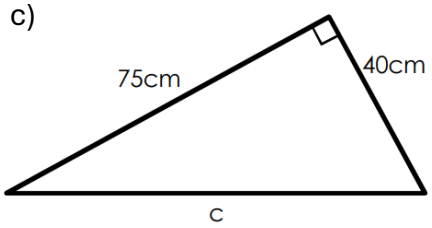
a)

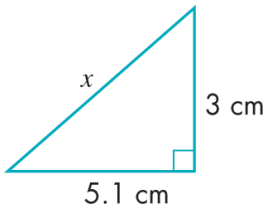
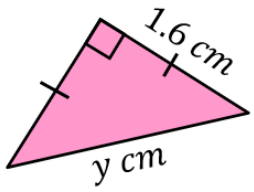
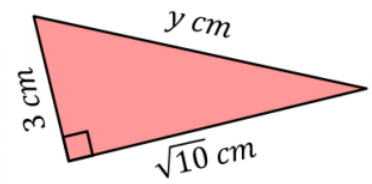


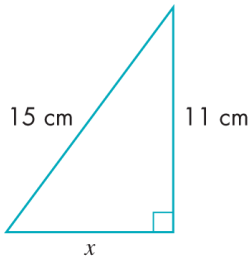
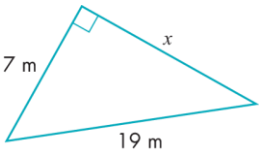
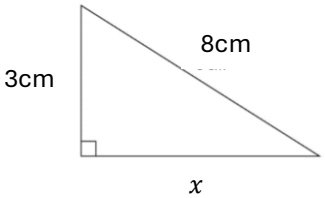
b)



c)

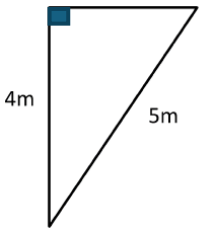
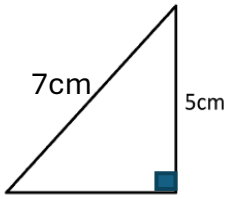
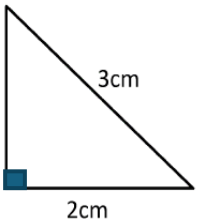


d)	e)	f)
		

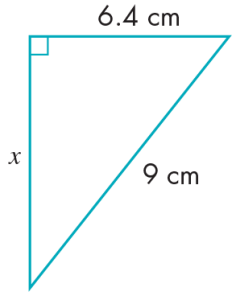
Steps	Use Pythagoras' theorem to find the missing side. Round to 1dp if necessary.		
	Example 2 I do...	***We do...***	***You do...***
1) Label the triangle (use c for the hypotenuse) 2) Substitute the values into Pythagoras' theorem $a^2 + b^2 = c^2$ 3) Rearrange to find the unknown. 4) Round your answer to the specified amount.			

Independent Practice

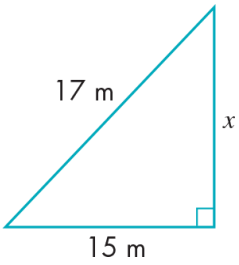
Question 2. Calculate the missing lengths and angles.

a)	b)	c)
		

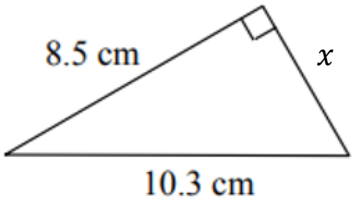
d)



e)



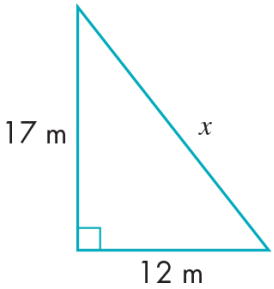
f)



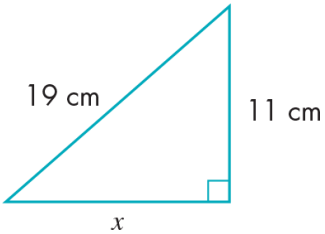
Mixed Practice

Question 3. Calculate the missing lengths.
Round your answers to 1 decimal place where necessary.

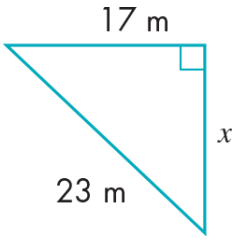
a)



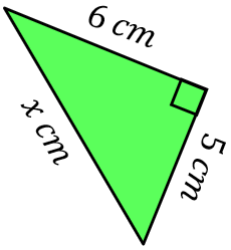
b)



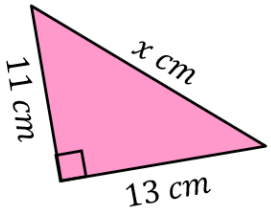
c)



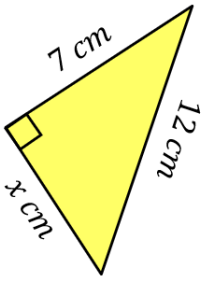
d)



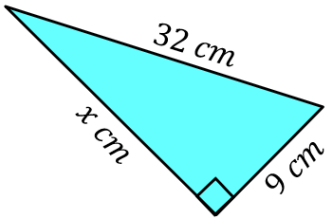
e)



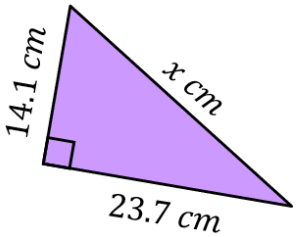
f)



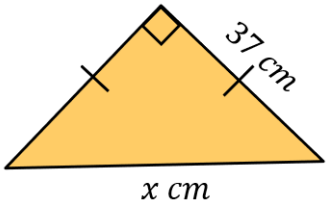
g)



h)

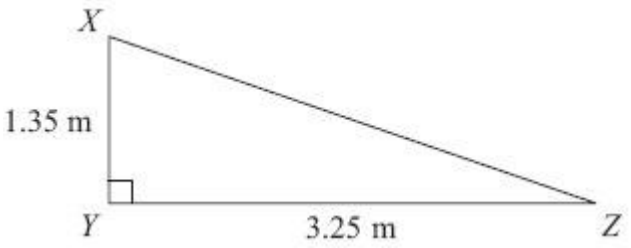


i)



Q1. *XYZ* is a right-angled triangle.

Calculate the length of *XZ*.
Give your answer correct to 3 significant figures.



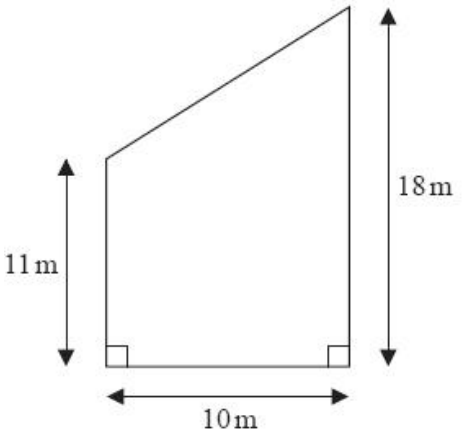
.....

(Total for Question is 3 marks)

Q2. Here is part of a field.
This part of the field is in the shape of a trapezium.
A farmer wants to put a fence all the way around the edge of this part of the field.

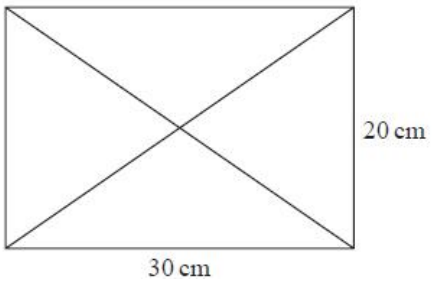
The farmer has 50m of fence.

Does he have enough fence?
You must show all your working.



(Total for question = 5 marks)

Q3. A frame is made from wire.
The frame is in the shape of a rectangle, 30 cm by 20 cm.
The two diagonals of the rectangle are also made from wire.
Calculate the total length of wire needed to make the frame and the diagonals.
Give your answer correct to 1 decimal place.



..... cm

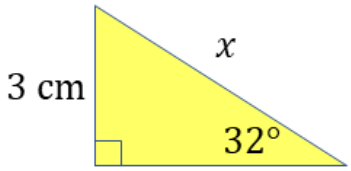
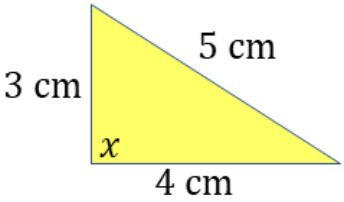
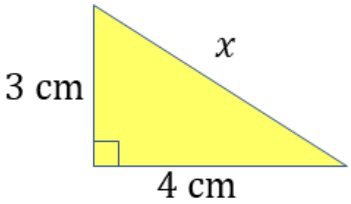
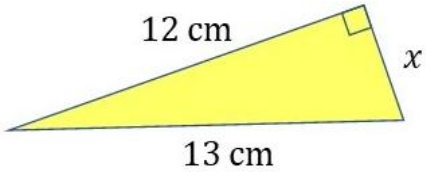
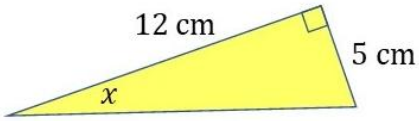
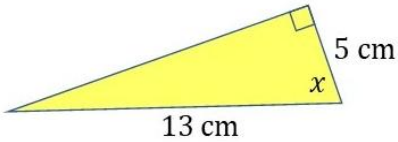
(Total for question = 4 marks)

Subtopic 2B: Choosing between Pythagoras and Trigonometry

Both Pythagoras' Theorem and Trigonometry can be used to solve problems involving right-angled triangles. Deciding which of these to use is a key skill for GCSE examinations:

What do you know?	What is the unknown?	Method
2 sides	A missing side	Pythagoras' Theorem
1 angle and 1 side	A missing side	Right-angled trigonometry
2 sides	A missing angle	Right-angled trigonometry

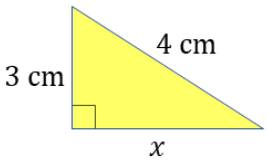
For each question, decide whether you would need to use Pythagoras' Theorem or right-angled trigonometry. **You do not need to calculate the unknown.**

Example 1 I do...	*We do...*	*You do...*
		
		

Independent Practice

Question 1. For each question, decide whether you would need to use Pythagoras' Theorem or right-angled trigonometry.

a)



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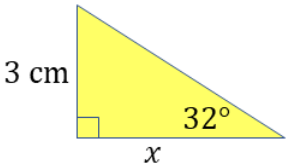

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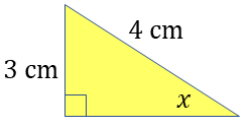

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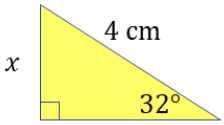

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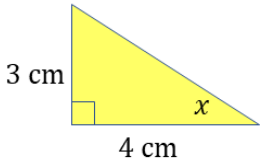

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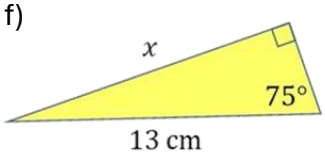

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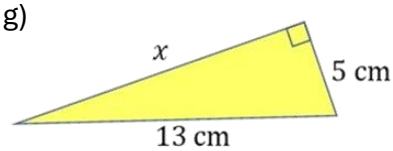

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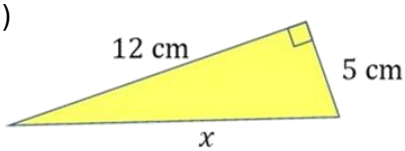

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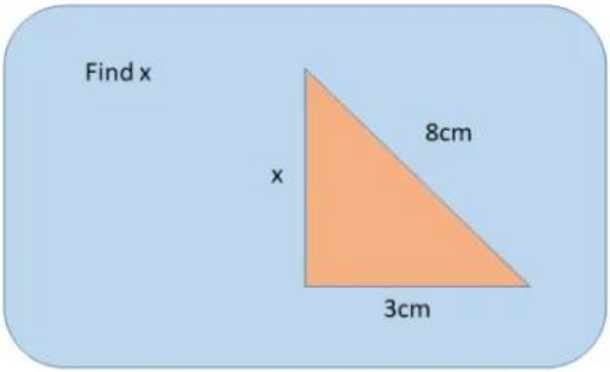
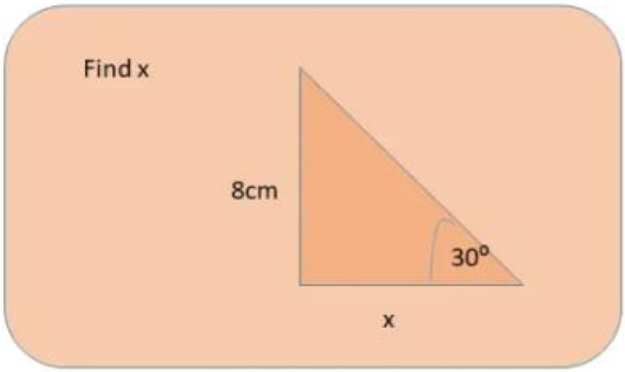
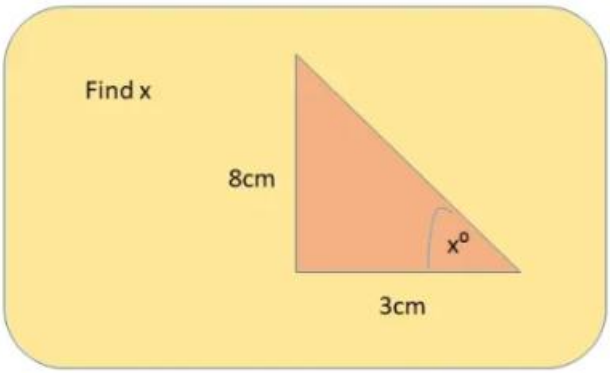
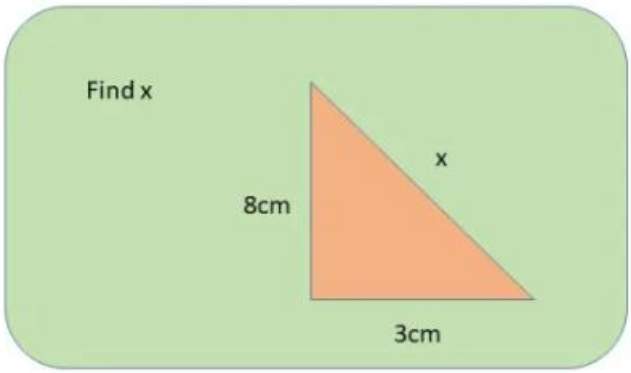
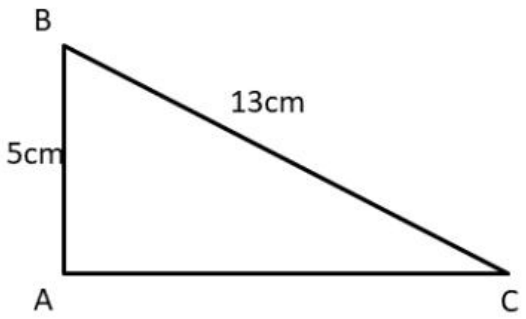
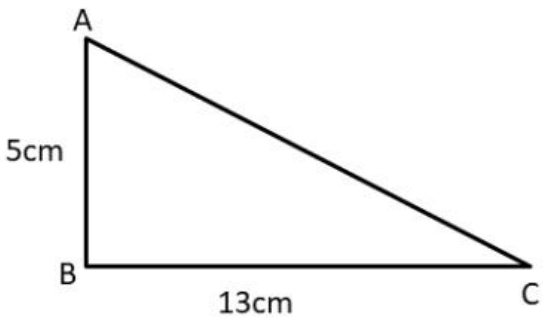




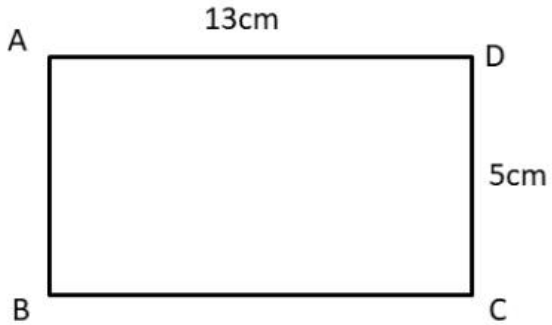


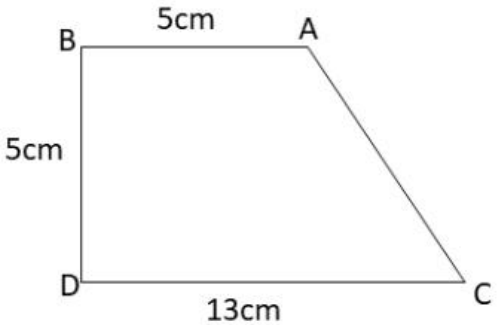






Find the length AC using 5cm and 13cm

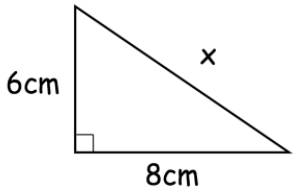




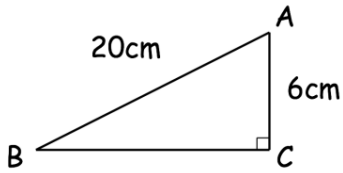
Method Selection

Mixed fluency

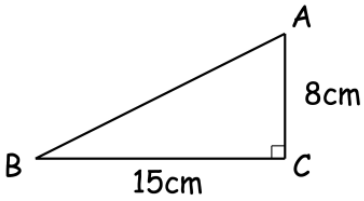
Q1.



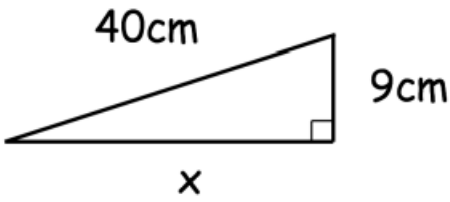
Q2.



Q3.



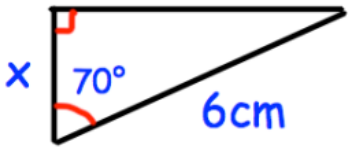
Q4. Charles says that the length of the missing side is 41cm. Here is his method:



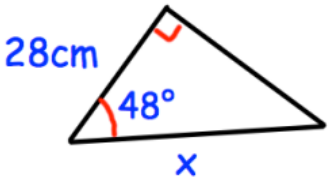
$$a^2 + b^2 = c^2$$
$$9^2 + 40^2 = x^2$$
$$81 + 1600 = x^2$$
$$x^2 = 1681$$
$$x = 41\text{cm}$$

Explain his mistake.

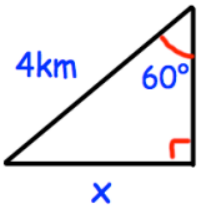
Q5.



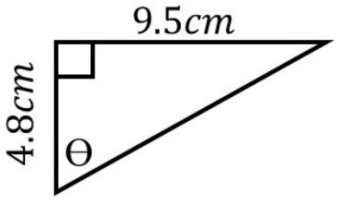
Q6.



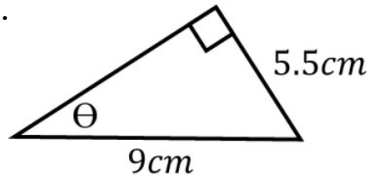
Q7.



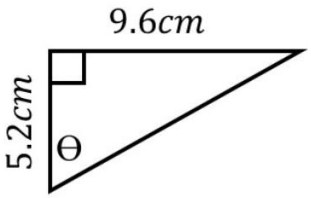
Q8

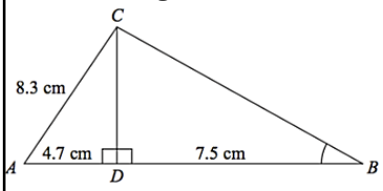
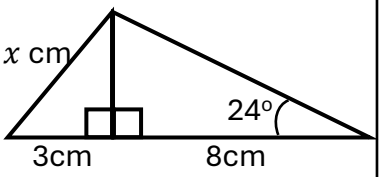
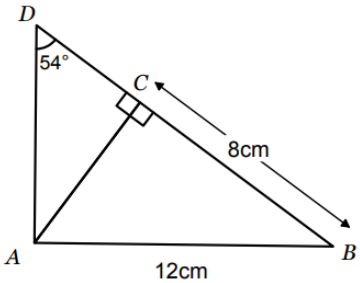


Q9.



Q10.

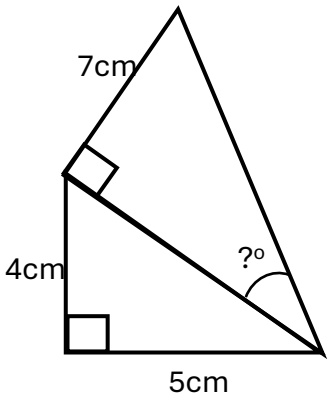


Steps	*Example 1 I do...*	*We do...*	*You do...*
1) Identify and highlight the shared side. 2) Calculate the value of the shared side by deciding whether to use Pythagoras' Theorem or trigonometry. 3) Use the value of the shared side to calculate x.	Calculate the size of angle ABC.  	Find the missing side length, x.  	Work out the length CD 

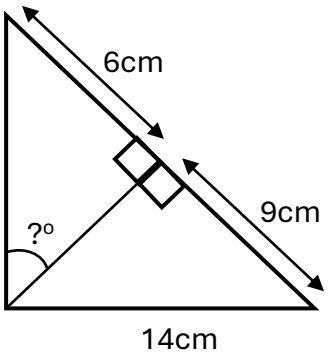
Independent Practice

Question 1. Calculate the value of the unknown.

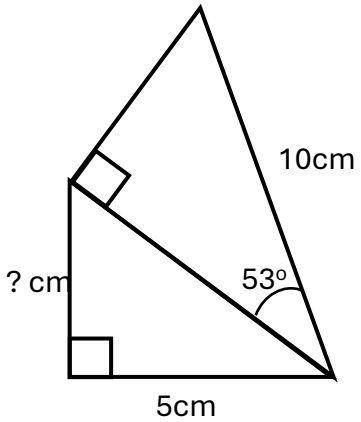
a)



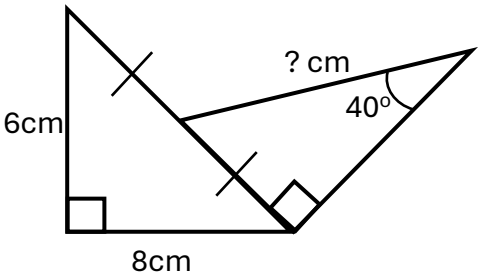
b)



c)

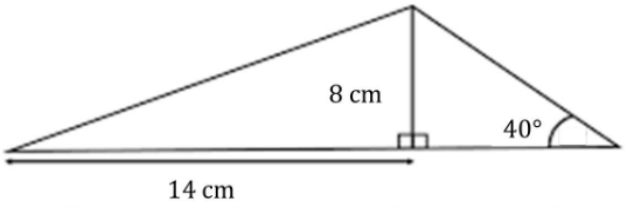


d)

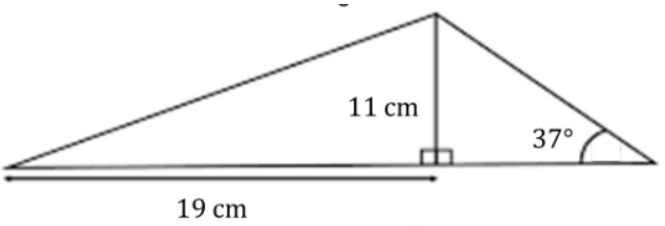


Question 2. Calculate the perimeter of the outer triangle:

a)

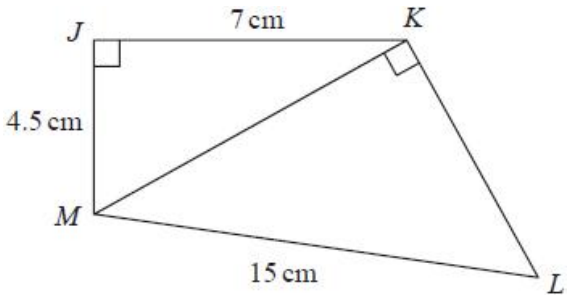


b)



Q1. The diagram shows a quadrilateral *JKLM*.

Work out the size of angle *KLM*.
Give your answer correct to 3 significant figures.



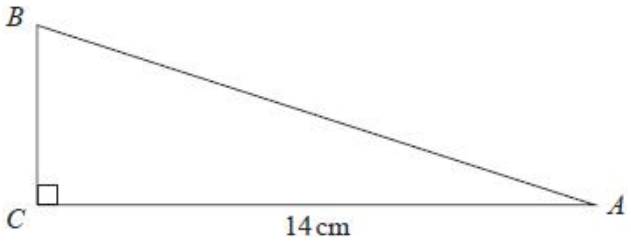
..... °

(Total for question = 4 marks)

Q2. ABC is a right-angled triangle.
AC = 14 cm.
Angle *C* = 90°

size of angle *B* : size of angle *A* = 3 : 2

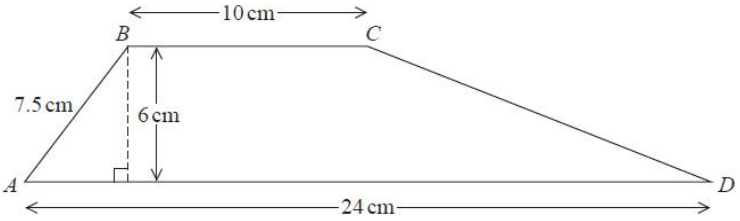
Work out the length of *AB*.
Give your answer correct to 3 significant figures.



..... cm

(Total for question = 4 marks)

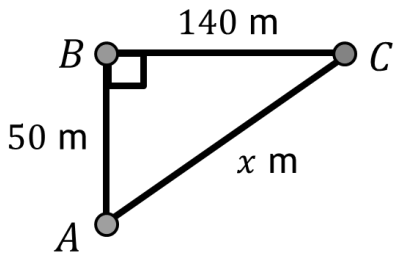
Q2. ABCD is a trapezium.
Work out the size of angle CDA.
Give your answer correct to 1 decimal place.

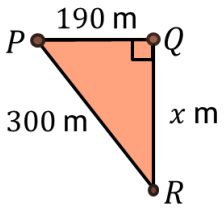
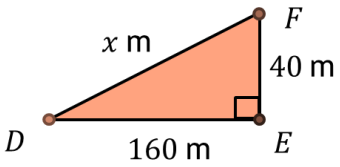


..... °

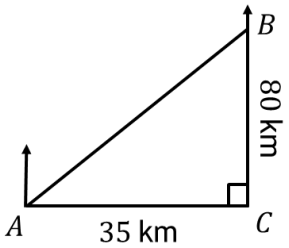
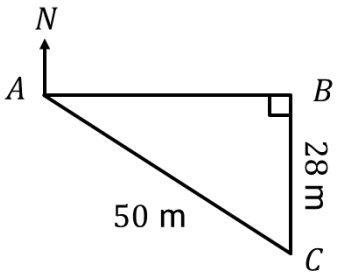
(Total for question = 5 marks)

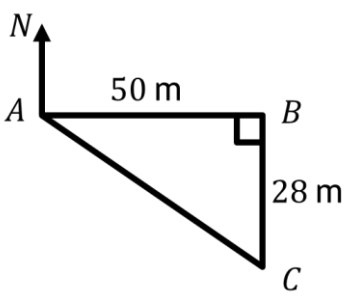
Using compass directions, we often create a right-angled triangle between points. This then allows us to solve more complex problems with bearings and direction using trigonometry and Pythagoras' Theorem.

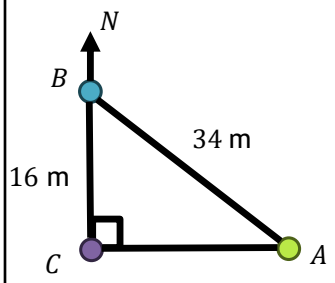
Steps	*Example 1 I do...*
1) If needed, draw a diagram of the bearings given. 2) Identify the right-angled triangle. 3) Calculate the missing lengths using Pythagoras' Theorem. $a^2 + b^2 = c^2$	From point A, Shula walks 50 m due North to point B From point B, she then walks 140 m due east to point C Work out the length of AC. Round your answer to 1 decimal place 

We do...	*You do...*
From point P, Abdul walks 190 m due east to point Q. From Q he then walks x m due south to point R. Abdul is now 300 m from point P. Find the value of x. Round your answer to 1 d.p.  	From point D, Amelie walks 160 m due east to point E. From E she then walks 40 m due north to point F. How far is Amelie from point D? Round your answer to 1 d.p. 

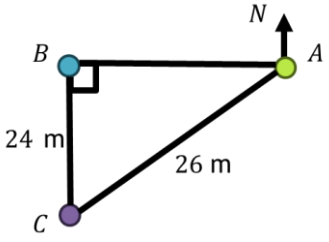
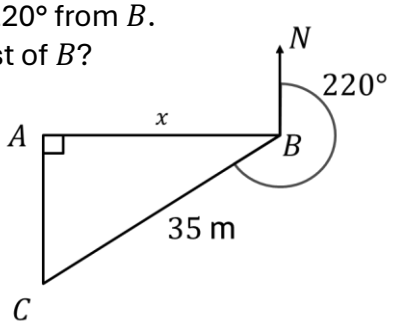
Independent Practice

Q1. Town B is 35 km east and 80 km north of town A. Work out the shortest distance from Town A to Town B. Give your answer correct to one decimal place.  	Anne and Bob each walk in different directions from Carl, as shown in the diagram. What is the distance between Anne and Bob rounded to one decimal place? 

Steps	Example 2 I do...
1) If needed, draw a diagram of the bearings given. 2) Identify the right-angled triangle & the required bearing. 3) Calculate the missing angles using trigonometry. 4) Calculate the bearing.	B is 50 m due east of A and 28 m due north of C Find the bearing of C from A 

We do...	You do...
The bearing of C from A is 200°. B is due south of A and due east of C. How far is C from A? (1 d.p.)	C is 16 km due south of B and due west of A. B is 34 km from A. Find the bearing of A from B. 

Exam-Style Questions

Q3. B is 24 km due north of C and due west of A. C is 26 km from A. Find the bearing of C from A.. 	**Q4.** A is due north of C and is due west of B. C is 35 m from B. C is on a bearing of 220° from B. How far is A due west of B? 

Subtopic 3: Exact Trigonometric Values

LEARNING OUTCOME: To solve trigonometry problems without a calculator.

Objectives:

1. To know exact trigonometric values for specific angles in a right-angled triangle.
2. To use exact trigonometric values to solve problems with right-angled triangles.

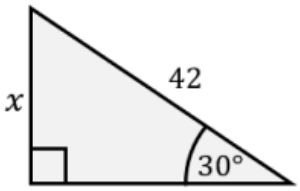
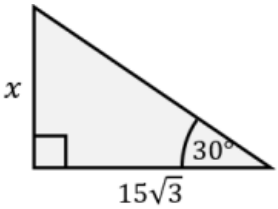
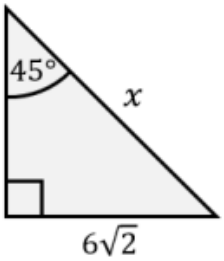
Keywords:

- Hypotenuse
- Opposite
- Adjacent
- Sine
- Cosine
- Tangent

There are several values for trigonometric ratios that you need to memorise for GCSE mathematics:

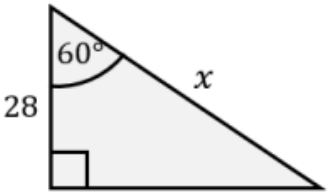
We can use these to solve problems involving trigonometry without a calculator.

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin(\theta)$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos(\theta)$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan(\theta)$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	n/a

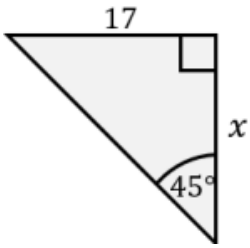
Steps	*Example 1 I do...*	*We do...*	*You do...*
1) Label the sides & choose the correct ratio. 2) Substitute in the known values, including the exact values. 3) Use equivalent fractions or rearrange and solve to find x.	Find the missing length.  	Find the missing length.  	Find the missing length. 

Independent Practice

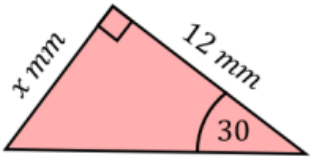
a)



b)



c)



Q1. Write down the exact value of cos 60°

.....

(Total for question = 1 mark)

Q2. Write down the value of sin 30°

.....

(Total for question = 1 mark)

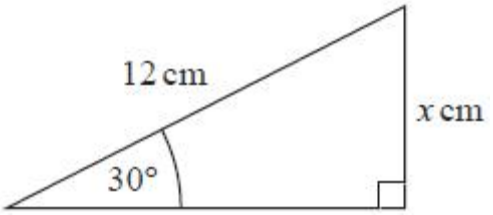
Q3.

(a) Write down the exact value of cos30°

.....

(1)

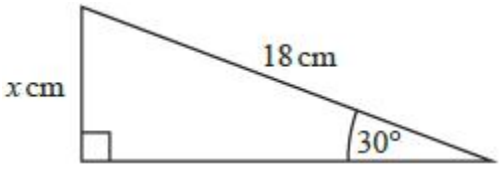
(b) Given that sin30° = 0.5,
work out the value of *x*.



.....

(2)

Q4. Work out the value of *x*.



x = ...

(Total for question = 2 marks)